8.5.2 CL:AIRE Development Industry Code of Practice

CL:AIRE (Contaminated Land: Applications in Real Environments) works with its members to raise awareness and pursue shared objectives in land, water and environmental management by collecting strategic industry information and developing initiatives that improve efficiency and save money. It has developed a Definition of Waste Code of Practice (DoW CoP) for dealing with the waste management aspects of contaminated land, which provides a clear, consistent and efficient process that enables the reuse of excavated materials on site, or their movement between sites. It has extended the scope significantly to further allow sustainable remediation and development of land, and to ensure that the industry uses materials sustainably within a robust technical and regulatory framework. Scenarios covered are shown below.

- Reuse of excavated materials on the site of production (contaminated and uncontaminated).
- Direct transfer of clean, naturally occurring soils between sites.
- Reuse of naturally elevated substances in soils (such as arsenic and lead).
- Cluster projects (multiple reuse at different development sites within a similar timeframe).
- Brownfield to brownfield transfers.
- Fixed soil treatment facilities allowing the release of treated materials to the market place.

The main purpose of the DoW CoP is to achieve good practice across the development industry to:

- assess whether materials are waste or not
- determine when treated waste ceases to become waste
- provide an auditable trail to demonstrate that the DoW CoP has been complied with on each site.

The DoW CoP specifies the implementation of a **materials management plan** (MMP), together with a declaration from a competent qualified person, before the commencement of the works. By complying with this DoW CoP, it may be possible to avoid the need to apply for a waste permitting exemption for the use of construction waste (U1). It also allows the direct transfer of uncontaminated natural excavation materials between projects without the need for a permit.

www **Further advice should be sought from CL:AIRE with reference to the DoW CoP and for further information on the register of materials and services.**

Flowcharts providing further guidance on how the CL:AIRE CoP links with environmental permitting in regard to the use or reuse of soils can be found in Chapter E10 Waste and material management.

CL:AIRE Code of Practice

Galliford Try – Ingsbeck flood alleviation scheme

Background

The voluntary DoW CoP developed by CL:AIRE in conjunction with the EA helps determine whether materials are classed as waste. The DoW CoP has recently been updated (CL:AIRE CoPv2) to allow the direct transfer of naturally-occurring soil materials.

Ingsbeck flood alleviation scheme is a £11 million development in Wakefield. It is spread across a number of areas and comprises the construction of new flood defence walls, channels, embankments and flood storage areas. Part of the construction involved building a new clay flood defence embankment around residential houses, which was valued at approximately £180,000. This required approximately 4,000 tonnes of clay. By applying the DoW CoP, a materials management plan (MMP) was produced to enable the reuse of this material, which had a number of benefits, shown below.

Reduced operational costs

A large volume of waste material would require the use of a standard rules environmental permit, which takes approximately four months in application and incurs costs of around £6,000 for application, subsistence and surrender, as well as the use of a technically competent manager. However, the MMP took three weeks from production to sign off and only cost £500. This benefited the project by reducing programme time and cost, which significantly decreased the overall project cost by £60,000, approximately 33% of the project value.

Reduced landfill costs

The surplus material would have been destined for landfill as there was no further use on the donor site. By utilising the MMP, 4,000 tonnes of material was diverted from landfill (avoiding a £10,000 landfill gate fee for importing inert material) and further benefiting from an 80 tonne embodied carbon saving.

Reduced use of natural resources

By utilising a recycled material, this avoided having to excavate a finite material from a quarry.

Reduced regulatory effort

The use of the MMP does not require any direct involvement from the EA and NRW regulator. This frees up its resources for deployment on other tasks and allows self-regulation for the industry, whilst minimising impact and protecting the environment.

8.5.3 Recycled aggregates

Recycled aggregates are materials developed from reprocessing of materials that were originally used in construction, including sand, gravel, concrete, stone, bricks asphalt, blast furnace and steel furnace slag, and, more recently, geosynthetic aggregates and recycled glass. Using recycled aggregates is cost effective, eco friendly and versatile. Benefits include lowering embodied energy and reducing transport if recycled on brownfield sites where the aggregates were produced.

The extraction of virgin aggregates has a wide range of impacts upon the environment, including noise, vibration, vehicle emissions, visual impact on the landscape and hydrogeology (the impact on the movement of groundwater in soil and rocks). Where a site with existing buildings is being redeveloped there is the potential to recycle materials from the buildings that are reaching their end of life status. Good planning can facilitate making them available as new construction materials or processing them into recycled aggregates.

On-site processing of demolition materials into aggregates is classed as a waste operation and will require an environmental permit or exemption for the treatment and reuse of the material. The WRAP (Waste and Resources Action Programme) quality protocol for the production of aggregates from inert waste *(refer to 8.5.3.1)* explains the requirements for ensuring that the processed materials meet end-of-waste status. You have a responsibility to check that all relevant supplier documentation is correct to confirm that the protocol requirements have been met or that the relevant permit or exemptions that may apply are in place.

www **For further information on the *Quality protocol: aggregates from inert waste* visit the Government website.**

For further details on reducing waste and permitting requirements associated with the treatment of demolition waste and contaminated soils refer to Chapter E10 Waste and material management.

8.5.3.1 WRAP quality protocol for the production of aggregates from inert waste

This protocol is published by WRAP and has been produced by the Quarry Products Association (QPA), the Highways Agency (HA) and WRAP as a formalised quality control procedure for the production of aggregates from recovered inert waste. These are referred to in the document as *Recovered aggregates*. The document has two main purposes.

1. To assist in identifying the point at which the inert waste used to produce recovered aggregates has been fully recovered, ceases to be a waste and becomes a product.
2. To give adequate assurance that recovered aggregate products conform to standards common to both recovered and primary aggregates.

You have a responsibility to check that all relevant documentation is correct to confirm that the protocol requirements have been met.

www **For further information visit the WRAP website.**

For flowcharts providing further guidance on how the WRAP quality protocol links with environmental permitting for the reuse of aggregates refer to Chapter E10 Waste and material management.

8.5.4 Reuse of excavated soils

Soil is a vulnerable and essentially a non-renewable resource because it can take more than 500 years to form a 2 cm thickness.

Topsoils contain living organisms that provide essential ecosystem services, supporting the production of food, and the creation of environments for wildlife and human recreational activity (such as parks and sports fields).

Soil management is increasingly seen as an essential part of any project. If planned and designed carefully soil can be re-incorporated into the project, avoiding the need to remove it from site. Even soils with contamination from other materials can be processed to make them suitable for reuse.

Re-incorporating soil into the project is an efficient use of resources

For further information refer to Chapter E09 Soil management and contamination control.

CONTENTS

Soil management and contamination control

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Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.

Scan the QR code to complete a brief survey, and share your feedback on the course.

Summary of soil management and contamination control legislation and guidance

This list is not exhaustive and only includes legislation mentioned in this section of GE700.

Legislation and guidance	Enforcement agencies*				
	EA	LA	NIEA	NRW	SEPA
Acts (primary legislation)					
Environment Act 2021	✓			✓	✓
Environmental Protection Act 1990	✓	✓	✓	✓	✓
Waste and Contaminated Land (Amendment) Act (Northern Ireland) 2011			✓		
Regulations (secondary legislation)					
Contaminated Land (England) Regulations 2006	✓				
Contaminated Land (Scotland) Regulations 2000					✓
Contaminated Land (Wales) Regulations 2006				✓	
Control of Asbestos (Amendment) Regulations (Northern Ireland) 2012			✓		
Control of Asbestos Regulations 2012	✓			✓	✓
Environmental Liability (Prevention and Remediation) (Amendment) Regulations (Northern Ireland) 2015			✓		
Environmental Liability (Scotland) Regulations 2009					✓
Radioactive Contaminated Land (Modification of Enactments) (England) Regulations 2006	✓				
Radioactive Contaminated Land (Modification of Enactments) (Wales) Regulations 2006				✓	
Radioactive Contaminated Land (Scotland) Regulations 2007					✓
Surface Waters (Fishlife) (Classification) (Scotland) Amendment Regulations 2003					✓
Guidance					
CL:AIRE CAR-SOIL™ *Asbestos in soil and construction and demolition materials*					
Code of Practice for the design of protective measures for methane and carbon dioxide ground cases for new buildings (BS 8485)					
Construction Code of Practice for the sustainable use of soils on construction sites (Defra)					
Investigation of potentially contaminated sites (BS 10175)					
Model procedures for the management of land contamination (CL:AIRE and Defra)					
Net Regs Environmental guidance					
The CL:AIRE *Definition of waste: Development industry Code of Practice*					
Organisations with information and guidance on their websites	✓	✓	✓	✓	✓

*Key

CL:AIRE	Contaminated Land: Applications in Real Environments
Defra	Department for Environment, Food and Rural Affairs
EA	Environment Agency
LA	Local Authorities
NIEA	Northern Ireland Environment Agency
NRW	Natural Resources Wales
SEPA	Scottish Environment Protection Agency

09

Overview

This chapter gives a general overview of the legal framework for the management of soil before and during construction and the definition, regulation and management of contaminated land.

The chapter identifies the processes involved with soil management and the assessment and remediation of contaminated land, the licences that will be required for its treatment or disposal and guidance for the prevention of pollution. It will also identify the main health and safety considerations when dealing with contaminated land.

9.1 Introduction

Building and construction works take place in many different environments but general descriptions of greenfield and brownfield are commonly used. Greenfield land (a site located in a rural area not previously built on) is more likely to require effective management of soils and subsoils due to the rural environment. Increasingly, though, the redevelopment of brownfield land, previously used for commercial or industrial activities, has grown in recognition of the importance of conserving our rural environment. Bringing these brownfield sites back into use is also considered a more sustainable option than using greenfield land.

The UK's extensive industrial past has, however, left a legacy of contamination on many sites and this needs to be managed to ensure the sites are suitable for their new purpose. Both the surface and the ground beneath may be contaminated by materials that have been worked, stored, spilt, buried, dumped or abandoned on the land in previous years. This list will also include the residue, waste or by-products from some industrial processes and the ashes from fires. Both solid and liquid waste may have permeated the ground to a considerable depth.

Sites that have had previous industrial occupation should be assumed to be polluted, and tests undertaken to ascertain the types of pollutant and their levels of concentration.

Everyone involved in work on both greenfield and brownfield land must be made aware of the importance of soil management and the possibility of contamination. They must make an assessment of the potential risks to the project and the negative impacts to human health and the environment, and implement management and mitigation processes.

Contaminated land

For the purpose of the regulations ***contaminated land*** is defined as any land that appears to be in such a condition, by reason of substances in, on or under the land, that significant harm or significant pollution of controlled waters is being caused or there is a significant possibility of such harm or pollution being caused.

9.2 Important points

- There are two main Codes of Practice that support soil and contaminated land management.
  1. The *Code of Practice for the sustainable use of soils on construction sites* is used to protect soils and ensure adequate soil function (for example, plant growth, water attenuation and biodiversity) during and after construction.
  2. The *Definition of waste: development industry Code of Practice* (DoW CoP) to provide a clear and concise process to determine whether excavated materials on a development site constitute waste in the first instance, and to identify the point when treated waste can no longer be considered as waste.
- Soils have a range of characteristics and it is important to understand the different properties and the advantages and disadvantages they offer with regards to drainage and growing mediums.
- Storage of soils must be carefully planned to avoid negative effects of compaction from stockpiles being too big and excessive traffic movements.
- Different classifications of soil should be stockpiled separately to avoid cross-contamination.
- The contract documentation and planning conditions for a project will identify known contamination of the site and the agreed methods for dealing with it. These should be referred to, and their requirements included, in the construction environmental management plan.
- Where a contaminated land assessment needs to be undertaken, this will be carried out in accordance with a systematic process of investigation, testing and appraisal to identify the most appropriate method for its treatment. The agreed method should be approved by the regulators before works commence.
- The testing of contaminated land must be carried out by certified, competent professionals in accordance with standard field testing and laboratory procedures approved by the regulators.
- All contaminated areas of the site must be fenced off and have clear exclusion signs to avoid unauthorised access and accidental spread of contamination across the site.

- The use of mobile plant for remediation of contaminated soils will require an environmental permit for the equipment and the need to complete a site deployment form detailing the work and management of the risks at each specific location.
- Where, following treatment, materials are still classed as waste, then their use will also require an environmental permit or registered exemption. The Contaminated Land: Applications in Real Environments (CL:AIRE) development industry Code of Practice can be used to ensure that contaminated materials are treated and managed so that they achieve an end of waste status. This will require the implementation of a materials management plan and a declaration by a qualified person that all requirements have been complied with.
- The stockpiling of contaminated soils should be avoided. Where stockpiling is unavoidable, material should be placed on impervious ground to avoid contaminates seeping into the ground and it should be covered to avoid wind-blown contamination and run-off to drainage systems and watercourses.
- Any removal of waste off site for treatment elsewhere or disposal should comply with the duty of care, including the provision of waste transfer or consignment documentation.
- All vehicles carrying contaminated materials off site should be appropriately sheeted and should pass through a wheel-wash facility to avoid contamination of the public highway.
- All personnel involved in the treatment of contaminated land should be aware of the relevant risks associated with the particular contaminants and wear the appropriate protective clothing, gloves and boots. Depending on the level of risk, a decontamination unit may need to be employed to prevent the spread of contaminants to clean areas of the site.

9.3 The importance of soil

Soil is vulnerable and because it can take more than 500 years to form a 2 cm thickness it is, in practical terms, non-renewable. One hectare of topsoil, the most productive soil layer, can contain up to 5 tonnes of living organisms. The importance of soil should not be overlooked. It is a resource that impacts on social, economic and environmental sustainability. It supports natural service to society, including the following.

- Food and fibre production.
- Environmental interaction (with water and air).
- Supporting ecological habitats and biodiversity.
- Supporting the landscape.
- Protecting cultural heritage.
- Providing raw materials.
- Providing a platform for construction.
- Supporting important eco-services (pollination and purification).

9.3.1 Planning for effective soil management

Soil, if planned and managed carefully, can be re-incorporated into the project, without the need to remove it from site because of poor design or contamination from other materials. Careful planning can help protect the quality and availability of topsoils. Make sure that you avoid the following construction activities that create negative impacts.

- Covering soil with impermeable materials, effectively sealing it and resulting in significant detrimental impacts on the soil's physical, chemical and biological properties, including drainage characteristics.
- Accidentally contaminating soil as a result of spillage or the use of chemicals.
- Over-compacting soil through the use of heavy machinery or the storage of construction materials.
- Reducing soil quality by mixing topsoil with subsoil.
- Mixing soil with construction waste or contaminated materials, which have to be treated before reuse or (as a last resort) sent to landfill.

The Department for Environment, Food and Rural Affairs (Defra), in support with the Waste and Resources Action Programme (WRAP), publishes a construction Code of Practice (CoP) for the sustainable use of soils on construction sites. Defra acknowledges that the publication has not been updated since 2009, but they still believe that it is a useful, relevant guidance document that can be used for reference purposes. Whilst the CoP is not legally binding, following it will help you to achieve the following.

- Protecting and enhancing soil resources on site and providing wider environmental benefits.
- Cost savings for your business.
- Supporting achievement of your business sustainability targets.
- Meeting your legal obligations regarding waste controls.

The CoP provides guidance on the various stages of site development where soil should be considered and contains ten sections to provide practical advice on different aspects of using soil sustainably on construction sites. Defra has partnered with CL:AIRE to review this document.

1. Knowing what soils are on site.
2. On-site soil management.
3. Topsoil stripping.
4. Subsoil stripping.
5. Soil stockpiling.
6. Soil placement.
7. Sourcing and importing topsoil.
8. Topsoil manufacture.
9. Soil aftercare.
10. Uses for surplus topsoil.

www **For further information on the *CoP for the sustainable use of soils on construction sites* visit the Defra website.**

9.3.2 Business benefits of good soil protection

Good soil management can identify opportunities for the reuse of topsoil on site, leading to cost savings, reuse of surplus subsoils, which might normally be sent to landfill, which in turn reduces the traffic movements to and from site to remove or import soils.

The identification of different types of soil will inform good storage processes and avoid cross-contamination between different soil types.

To facilitate the actions above, a soil resource plan should be undertaken.

The soil resource plan will set out in detail the methods, equipment, location, volumes and programme for the recovery, storage and reuse of all site topsoil and subsoil.

9.3.3 Protection measures for soil during construction

- Prepare a soil resource plan showing the areas and type of topsoil and subsoil to be stripped, haul routes, methods to be used, the location, type and management of each soil stockpile and named individuals responsible for soil management on site.
- When stripping, stockpiling or placing soil, do so in the driest conditions possible and use tracked equipment where possible to reduce compaction. If there is a sustained period of heavy rainfall (for example, >10 mm in 24 hours) stop work and only restart after the ground has had one full drying day or agreed criteria for the soil composition has been met.
- Confine traffic movement to designated routes.
- Keep soil storage periods as short as possible.
- Clearly define stockpiles of different soil materials.

Topsoils come with a wide range of different characteristics making them suitable for different landscapes and plants (for example, turfing, tree pits, wildflowers and grassland).

To ensure that the topsoil is suitable for the intended purpose, it is important to have the soil independently assessed against a topsoil specification.

The most functional specifications are those that list which properties the topsoil should possess prior to planting, turfing or seeding.

These normally include the following.

- Visual examination (for example, soil structure, consistency and foreign matter).
- Particle size analysis (texture) and stone content.
- pH and salinity values.
- Content of major plant nutrients.
- Organic matter content.
- Maximum levels of potential contaminants (for example, heavy metals, hydrocarbons, cyanide and phenols).

9.4 Regulatory bodies involved in contaminated land

Various organisations (as listed below) will be involved in granting approval for the treatment and the redevelopment of contaminated sites.

- The Local Authority, which has a statutory duty to inspect its land and identify any sites that are formally designated as contaminated.
- The Local Authority, before granting planning permission, will approve the remediation strategy to ensure that the ground is suitable for the proposed development.
- The Environment Agency (EA) for England, Natural Resources Wales (NRW) for Wales, the Northern Ireland Environment Agency (NIEA) for Northern Ireland and the Scottish Environment Protection Agency (SEPA) for Scotland, for issuing an environmental permit for the use of mobile treatment plant to treat contaminated soils or for the reuse of construction and demolition waste.
- The Local Authority for issuing an environmental permit for crushing equipment and for granting an exemption for the crushing and screening of demolition materials.
- The relevant environment agency for regulating the disposal of waste under the duty of care.
- The relevant environment agency for regulating contaminated sites that are deemed to be special sites (for example, areas that could seriously damage surface or groundwater supplies, or defence sites or radioactive sites).
- The Local Authority environmental health officers for dealing with any complaints regarding dust that crosses the site boundary.
- The water companies and the relevant environment agency for the disposal of polluted water from contaminated sites.

Effective contact with each of these authorities is essential, and must be established early in the project.

Any contaminated site must be totally fenced off and adequate warning notices must be prominently posted, advising all members of the public that the site is dangerous and to refrain from entering.

For further information on waste management refer to Chapter E10 Waste and material management.

9.5 Contaminated Land Regulations

Part IIA of the Environmental Protection Act 1990 (EPA) provides the legal framework for dealing with contaminated land in the UK.

It is implemented in each of the devolved administrations through the following regulations.

9.5.1 England

- Contaminated Land (England) Regulations 2006.
- Radioactive Contaminated Land (Modification of Enactments) (England) Regulations, which extend controls on contaminated land to radioactive contaminated land.

9.5.2 Northern Ireland

- The Contaminated Land Regime, which is set out in Part 3 of the Waste and Contaminated Land (Northern Ireland) Order 1997.

This regime is very similar to that provided in Part IIA of the EPA in England, Scotland and Wales. The District Councils will be the primary regulators for the regime. Initial activity will focus on the preparation of site inspection strategies, which are to be completed within 12 months of the regime being in place.

9.5.3 Scotland

- The Contaminated Land (Scotland) Regulations 2000, SSI 2007/178, and statutory guidance were brought into force in 2000 and are similar to the regulations that apply in England and Wales.
- Radioactive Contaminated Land (Scotland) Regulations 2007.

9.5.4 Wales

- Contaminated Land (Wales) Regulations 2006.
- Radioactive Contaminated Land (Modification of Enactments) (Wales) Regulations 2006, which extend controls on contaminated land to radioactive contaminated land.

The Local Authority (or relevant environment agency in the case of a special site) has a statutory duty to ensure the remediation of contaminated land is paid for by an appropriate person(s).

The appropriate person(s) will be the person(s) who knowingly permitted the pollution or, if they cannot be found, the responsibility will fall to the owner or occupier of the site.

To enforce the remediation works the regulatory authority will serve a remediation notice on the appropriate person(s) specifying what needs to be done and by when.

There is a close relationship between the contaminated land regime under EPA Part IIA above and planning controls.

The National Planning Policy Framework confirms that planning policies and decisions should ensure the following.

- The site is suitable for its proposed use, taking account of ground conditions and any risks arising from land instability and contamination. This includes risks arising from natural hazards or former activities (such as mining) and any proposals for mitigation including land remediation (as well as potential impacts on the natural environment arising from that remediation).
- After remediation, as a minimum, land should not be capable of being determined as contaminated land under Part IIA of the EPA.
- Adequate site investigation information, prepared by a competent person, is presented.

Planning conditions should ensure appropriate investigation, remediation, monitoring and record keeping. Where contamination is suspected, the developer is responsible for investigating the land to determine what remedial measures are necessary and the actual remediation work to ensure its safety and suitability for its intended purpose. There is a significant emphasis on voluntary remediation by the developer to avoid a formal remediation notice being issued by the regulating authority.

Where there is a requirement to treat or dispose of contaminated material, then waste controls and the duty of care will apply *(refer to 9.7 Managing contaminated land).*

9.6 Brownfield sites

The definition of a brownfield site generally relates to land that has had some form of previous development.

Many brownfield sites will be land that is affected by contamination but not to an extent that it automatically falls within the definition set out in the different versions of the Contaminated Land Regulations above. Previously published estimates of the extent of land affected by contamination vary widely, from 50,000 to 300,000 hectares, amounting to as many as 100,000 sites.

The Environment Agency estimates that, of these, 5,000 to 20,000 may be expected to be problem sites that require action to ensure that unacceptable risks to human health and the environment are avoided. Some brownfield sites are affected by land contamination because of the previous industrial uses of the site, which has led to the deliberate or accidental release of chemicals onto the land.

The following are examples of chemicals associated with four industrial processes.

Oil refineries (fuel, oil, lubricants, bitumen, alcohols, organic acids, polychlorinated biphenyls (PCBs), cyanides, sulphur and vanadium).

Lead works (lead, arsenic, cadmium, sulphides, sulphates, chlorides, sulphuric acid and sodium hydroxide).

Pesticide manufacturing (dichloromethane, fluorobenzene, acetone, methanol, benzene, arsenic, copper sulphate and thallium).

Textile and dye works (aluminium, cadmium, mercury, bromides, fluorides, ammonium salts, trichloroethene and polyvinyl chloride).

Sites with previous industrial occupation should be assumed to be polluted

9.7 Managing contaminated land

Managing land affected by contamination involves the identification of risks and then putting in the appropriate control measures to reduce those risks to an acceptable level so that the land is suitable for its intended use. Dealing with land contamination and regeneration helps towards making the environment clean and safe, enhances individual's health and wellbeing, and improves the economic, ecological and amenity value of the area.

The Environment Agency expects sites to follow the *Land contamination risk management* (LCRM) guidance when managing the risks associated with land contamination (applicable to England, Northern Ireland and Wales). This guidance can be used in Wales, but must also be used in conjunction with *Land contamination: our role in managing and dealing with land contamination* for differing requirements. For Northern Ireland, reference must also be made to the Department of Agriculture, Environment and Rural Affairs *Practice guide – redeveloping land affected by contamination*. Local authorities and other regulators may also provide additional guidance.

The LCRM can be used in a range of regulatory and management contexts, and helps with:

- identification and assessment of unacceptable risk.
- assessment of suitable remediation options to manage the risk.
- planning and carrying out remediation works.
- verifying that the remediation has been successful.

The LCRM consists of four guides:

1. LCRM: Before you start.
2. LCRM: Risk assessment.
3. LCRM: Options appraisal.
4. LCRM: Remediation and verification.

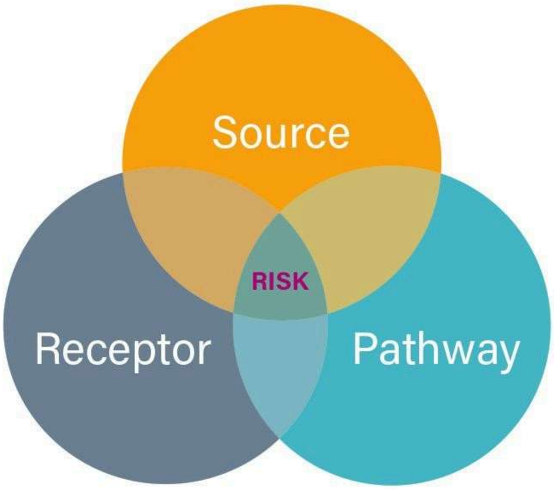
A staged risk-based approach is used, consisting of three stages. Each is broken down into tiers or steps.

www **For further information on the LCRM guidance, visit the Government website.**

9.7.1 Stage 1: Risk assessment

This stage has three tiers that are applied to approaching a risk assessment:

1. **Preliminary risk assessments** are the first tier of risk assessment, and develop the initial conceptual site model to establish whether there are any potentially unacceptable risks.
2. **Generic quantitative risk assessments** use generic assessment criteria and assumptions to estimate risk.
3. **Detailed quantitative risk assessments** use detailed, site-specific information to estimate risk (this stage also includes information for intrusive site investigations).



Land contamination risk assessments can only be completed a competent person. A technical approach to risk assessment must be followed for each tier of risk assessment, as follows:

- Identify the hazard – establish contaminant sources.
- Assess the hazard – use a source-pathway-receptor (S-P-R) linkage approach to find out if there is the potential for unacceptable risk.
- Estimate the risk – predict what degree of harm or pollution might result and how likely it is to occur by using the tiered approach to risk assessment.
- Evaluate the risk – decide whether a risk is unacceptable.

Pathways will be specific to the receptor type. For example:

- ingestion, inhalation or dermal contact for human health receptors
- infiltration and contaminant migration through permeable strata, such as the unsaturated zone for groundwater
- a secondary pathway from groundwater contamination to surface water
- migration of ground gases and vapours such as permanent gases, landfill gas and volatile hydrocarbons into buildings
- direct contact and uptake by plants.

You will develop an initial conceptual site model in the preliminary risk assessment that will need to be refined and updated as you progress through the LCRM.

Risk assessment evaluation criteria are the parameters used to judge whether particular harm or pollution needs further assessment or is unacceptable. Any evaluation criteria used must be:

- relevant to the site
- relate to the contaminants and receptors you are dealing with.

The exact choice of evaluation criteria will depend on:

- the reasons for doing the risk assessment and the regulatory context
- the conceptual site model and potential contaminant linkages
- any criteria set by regulators
- any advisory requirements such as from Public Health England
- the degree of confidence and precaution required to judge whether a risk is unacceptable
- how you have used or developed more detailed assessment criteria in further tiers of risk assessment
- the availability of robust scientific data
- how much is known – for example, about the pathway and how the contaminants affect receptors
- any practical reasons, such as being able to measure or predict against the criteria.

9.7.1.1 Evaluate the risks

For preliminary risk assessment, the risks need to be evaluated qualitatively. If you progress to a generic or detailed quantitative risk assessment, use evaluation criteria to judge whether particular harm or pollution needs further assessment or is unacceptable. You will need to collect detailed investigation data to support the use of evaluation criteria.

www **For information on each of the risk assessment tiers, refer to the Government website.**

9.7.2 Stage 2: Options appraisal

You must be a competent person to conduct an options appraisal. If this stage has been reached, then it has been decided that there is a need to remediate. Remediation is the action required to prevent, minimise, remedy, or mitigate the effects of the unacceptable risks.

This stage allows you to identify an appropriate remediation option, by following three steps:

1. Identify feasible remediation options.
2. Undertake a detailed evaluation of options.
3. Select the final remediation option.

9.7.2.1 Step 1: Identify feasible remediation options

To identify possible remediation options for further evaluation, the following needs to be considered:

- Is the information available sufficient?
- Formulating the appraisal objectives of options.
- Producing a shortlist of potential remediation options.
- Selecting the most feasible options, to assess in more detail.

When considering options, any required regulatory controls, such as permits, must be factored in. Regulatory controls covering most land contamination treatment activities will use one of the following:

- standard rules: SR2008 No 27: *Mobile plant for the treatment of waste soils and contaminated material, substances or products* and Form MPP2: *deployment form for land and groundwater remediation*
- a bespoke mobile plant permit for the treatment of waste soils and contaminated material, substances, or products
- a site-based permit.

Ahead of remediation, plan for any requirements such as for planning and Part 2A. If you intend to use the CL:AIRE Definition of waste: development industry code of practice (DoW CoP) you can find further details on the CL:AIRE website.

If you intend to use the CL:AIRE Definition of waste: development industry code of practice (DoW CoP) you can find further details on the CL:AIRE website.

At the end of step 1, the following will have been concluded:

- identification of options appraisal management and technical objectives
- specific remediation objectives and criteria
- consideration of the regulatory controls that may be required
- production of a shortlist of feasible remediation options that will be further evaluated during step 2.

For more detailed information on step 1, visit the Government website.

9.7.2.2 Step 2: Do a detailed evaluation of options

This step decides which options will be the most suitable for dealing with each relevant contaminant linkage. To achieve this, you will need to:

- assess the limitations, advantages, and disadvantages of each option
- develop and use options appraisal evaluation criteria to assess the merits of each option
- establish which options are most suitable – individually or combined
- put forward proposals for combining options
- ensure that detailed information on the technical aspects of each option, including the cost, are sourced.

The level of detail required for the evaluation criteria will be dependent upon site circumstances. The evaluation used needs to take account of the best practicable environmental option. Examples of criteria that must be considered when selecting an option(s) include the following.

Regulatory and stakeholder requirements. The option selected must be acceptable to relevant stakeholders, such as the regulator, client, site purchaser and local residents.

Sustainability. To address sustainability, Stage B of the SuRF-UK Framework can be adopted. This can be used to differentiate the remediation options against environmental, social, and economic indicators considering any relevant climate change issues. You can also refer to BS ISO 18504: Soil quality. Sustainable remediation.

Cost. Affordability needs to be assessed in line with the available resources and the approximate cost of verification and remediation.

Timescales. Consideration of how long it will take to complete remediation, the need for any post remedial monitoring and any future obligations required such as post-remediation or long-term monitoring and maintenance.

Practicability. Is the option practical regarding the site location, layout, size, maintenance needs, operational needs, compatibility with other planned site works.

Effectiveness. This needs to address if the remediation will successfully reduce or control the unacceptable risks to a pre-determined level, that the timescales are realistic, the option is applicable to the relevant contaminant linkages and if effectiveness can be demonstrated throughout the verification process.

Durability. This needs to address if the remediation will successfully reduce or control the unacceptable risks over a certain time.

Environmental impact. Will the remediation deliver direct and indirect benefits and how will it affect the quality of the environment during and after completion.

Track record. Identify evidence of other successful remediation using your selected approach.

Availability. Are the methods or techniques available and is there a need for specialist contractors or specific reagents.

Health and safety requirements. What levels of health and safety are needed associated with each option.

For more information on the technical basis of selected remediation methods, refer to:

- **INFO-OA2: detailed evaluation of remediation options on the CL:AIRE website**
- **Contaminated land guidance tool for references on the CIRIA website**

At the end of step 2, the following will have taken place:

- an assessment of which remediation options are the most feasible for each relevant contaminant linkage
- a decision will have been made on any options that can or need to be combined.

For more detailed information on step 2, visit the Government website.

9.7.2.3 Step 3: Select the final remediation option

This stage focuses on which option is the most suitable, which could be a single or combined approach. The final option must meet the overall site objectives identified in the preliminary risk assessment, the options appraisal management and technical objectives and the options appraisal remediation objectives.

It may not be easy to identify a remediation option that meets some or all the remediation objectives completely. Some of the reasons for this could be due to uncertainty on reducing or controlling the risks, achievability within the requirement timeframe, not practical because of the sites size, location, access, or topography, or it is too expensive despite it being the most effective, practicable and robust solution.

Stakeholders could have varying views on what is required, for example:

- the regulator, to meet regulatory and legal requirements, so this is most likely to be an important consideration
- the site owners' views on what they consider sufficient
- neighbouring property owners' views.

It may be necessary to review the remediation objectives, or alternatively:

- agreeing a lower standard of remediation, an example of this would be changing the layout or use of the site
- adjusting in other areas, such as the provision of additional health and safety protection
- looking at alternative remediation options such as new technology or approaches and use
- carrying out longer term monitoring and then re-evaluating, particularly if the site has complex contamination.

At the end of step 3, the following will have been achieved:

- selection of the final remediation option
- assessment of how a combined or integrated approach will work in practice
- a decision on actions needed if feasible options could not be identified
- a decision and agreement on longer-term monitoring and maintenance, or monitored natural attenuation as a remediation option
- confirmation that the selected remediation option will manage the risk effectively, and be verifiable at stage 3 of LCRM
- recording and justification of the decision in options appraisal report.

www **For more detailed information on step 3, visit the Government website.**

9.7.3 Stage 3: remediation and verification

This stage has four steps that need to be followed:

1. Develop a remediation strategy.
2. Remediate.
3. Produce a verification report.
4. Do long-term monitoring and maintenance, if required.

Only a competent person with training, knowledge and experience in remediation can complete stage 3.

9.7.3.1 Step 1: Develop a remediation strategy

A remediation plan will need to be developed and agreed. Monitoring objectives and criteria to track performance of the remediation will need to be established. This strategy will provide a record of how the remediation objectives will be met and conducted.

As part of the remediation strategy, a verification plan will need to be produced. This will contain all of the data requirements, including compliance criteria and monitoring details. This will establish an audit trail to verify remediation is working or has worked.

Once the remediation strategy has been agreed, and the correct regulatory controls are in place, the remediation can commence. A key part of this process will be remediation progress reporting and monitoring/maintenance reporting.

Once remediation is complete, a verification report is produced that demonstrates that the risk has been reduced and that the remediation objectives and criteria have been met. The verification report will detail a complete record of all remediation activities and evidence that it has been a success.

The remediation strategy and verification report are an integral part of the final record of the land quality.

www **For more detailed information on the remediation strategy, see INFO-OA3: developing the remediation strategy on the CL:AIRE Water and Land Library.**

9.8 Occupational health considerations

The health of workers on contaminated sites can be affected through one or more of the following ways.

- Asphyxiation.
- Gassing.
- Ingestion.
- Inhalation.
- Skin absorption.
- Skin penetration.

Under the Personal Protective Equipment at Work (Amendment) Regulations 2022, employers must carry out an assessment to ensure that the correct personal protective (PPE) clothing and respiratory protective equipment (RPE) are issued and worn at all times when work is carried out on contaminated sites.

Continuous assessments of the risk to health by exposure to any contaminated material or land must be carried out, and the control measures or precautions constantly monitored.

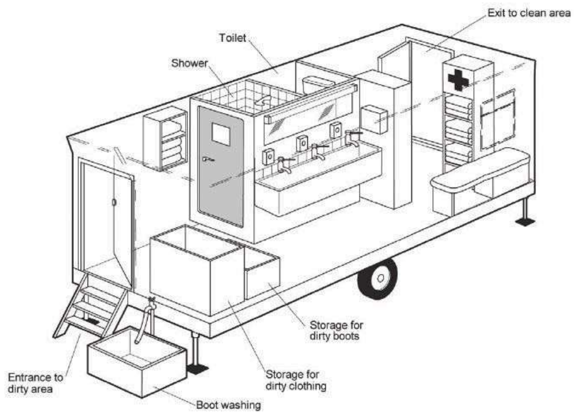
9.8.1 Personal hygiene

The level of risk to health by any contaminants will determine the need and scale of hygiene facilities, but certain consideration should always be borne in mind when working on a contaminated site.

A dirty area is required for workers to discard dirty or contaminated clothing. Such clothing should be bagged and identified within this area before being dispatched to specialist cleaners.

Washing and toilet areas. Toilets, showers and washing facilities should be positioned between the dirty and clean areas, so that workers may wash or shower in order to remove any contaminant from their bodies.

A clean area is required for workers to put on clean and non-contaminated clothing. Access to and exit from this area must only be to the clean part of the site. It is essential that the entry and exit point of the clean area is in the clean part of the site.



Typical layout of a hygiene unit, divided into three areas, with the dirty entrance remote from the clean exit

Daily cleaning of the toilet facilities and the decontamination of all facilities must be carried out.

9.8.2 Asbestos

Asbestos was widely used in construction and insulation materials and poor management and waste practices have led to asbestos contaminating brownfield sites. In addition, many unprotected sites have suffered from the indiscriminate fly-tipping of asbestos waste.

Asbestos is a highly dangerous material and presents substantial risks to the health of those who work with it and those who may come into contact with it. Asbestos-related diseases are responsible for around 5,000 deaths a year in Great Britain. Works dealing with asbestos come within the requirements of the Control of Asbestos Regulations 2012.

Before the commencement of works, the site assessment should identify any risks posed by asbestos and the appropriate mitigation actions for dealing with it, which may range from physical containment to disposal.

For further information on the management of asbestos waste refer to Chapter E10 Waste and material management.

CL:AIRE published the Joint Industry Working Group (JIWG) Asbestos in Soil and Construction & Demolition (C&D) materials guidance *Control of Asbestos Regulations 2012: Interpretation for Managing and Working with Asbestos in Soil and Construction* and *Demolition Materials: Industry Guidance*. Generally referred to as CAR-SOIL, this guidance document highlights the definitive explanation of how the legal requirements of the Control of Asbestos Regulations have been interpreted to apply to work with asbestos-contaminated soil, and construction and demolition materials. The aims of CAR-SOIL are listed below and overleaf.

- Bring together the asbestos management, occupational hygiene and brownfield management sectors with the aim of promoting the development of a consistent and harmonised approach to the regulation, investigation, analysis, assessment and management of asbestos in soil.
- Develop practical practitioner guidance on asbestos in soil that provides a consistent approach for UK industry, stakeholders and regulators.

- Promote the development of the relevant industry professional qualification framework for asbestos in soils for the brownfield and asbestos management sectors, building on the existing professional qualification framework for the management of asbestos in buildings, and relevant statutory requirements.
- Engage with the principal regulatory bodies for asbestos (the Health and Safety Executive (HSE), Environment Agency and representatives of Local Authority contaminated land officers) with the aim of promoting a consistent, unified and transparent regime for the regulation of all aspects relating to the remediation of land contaminated by asbestos.
- Promote and develop an improvement in public and stakeholder awareness of relevant issues, including health and safety, public health, technical, legal and insurance, related to the occurrence of and investigation and remediation of asbestos in soil.
- Promote the work of the Asbestos in soil JIWG to ensure all organisations are fully informed of its activities.

www **For further information about CAR-SOIL, visit the CL:AIRE website.**

9.8.3 Explosives

Extreme care must be taken on sites where explosives are known to have been stored or used. This includes old mine workings, coal mines, former explosives factories and Ministry of Defence establishments. Furthermore, unexploded bombs are occasionally unearthed when construction work takes place in areas that were subjected to bombing during World War II. Disturbing any explosives could have sudden and disastrous consequences, especially if they are old and starting to decay.

Once it is agreed that excavation work should proceed, this should be done with utmost caution. Any areas of soil discolouration, unusual objects or unusual cable presence should be taken as an indication that explosives are present. Work should be stopped immediately and the police informed.

Well-established procedures already exist for competent military personnel to deal with unexploded devices.

9.8.4 Anthrax

Anthrax spores may lie dormant within soil, or in horse or cow hair binders in old lath and plaster, for many decades.

When such spores are disturbed, they still have the capacity to cause severe environmental problems. You should regard all premises (such as old tanneries, wool sorting stations and premises used in connection with animal carcasses, hides, bones, offal, or for the production of gelatine and old lath and plaster walls or ceilings) as high-risk areas where anthrax spores may be present.

Defra will be able to supply advice as to whether contaminated carcasses have been buried on old farm sites.

Where it is suspected that anthrax spores may be present, it is essential for everyone on the site to exercise good personal hygiene and use impervious personal protective clothing, including gloves. Any cuts and scratches must be adequately covered. As an additional safeguard, advice on immunisation and general health procedures should be sought from a doctor.

9.8.5 Radiation

Before starting work on a site where any work involving radioactive materials has previously taken place (such as hospitals) or where radioactive contamination (whether natural or artificial) may be present, consult the HSE and the relevant environment agency.

It will be necessary to use specialist contractors for all aspects of both the removal of substances and decontamination where radioactive materials are being dealt with.

9.8.6 Persistent organic pollutants

Persistent organic pollutants (POPs) are poisonous chemical substances that break down slowly in the environment, bioaccumulate (build up in the body or in organisms over time) through the food chain, and pose a risk of causing adverse effects to human health and the environment. POPs can be transported across international boundaries, far from their sources, even to regions where they have never been used or produced. The manufacturer, sale and use of products containing POPs is now banned.

The ecosystems and indigenous people of the Arctic are particularly at risk because of the long-range environmental transportation and bio-magnification of these substances. Consequently, POPs pose a threat to the environment and to human health all over the globe.

CONTENTS

Waste and material management

10

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.

Summary of waste and material management legislation and guidance

This list is not exhaustive and only includes legislation mentioned in this section of GE700.

Legislation and guidance	Enforcement agencies*				
	EA	LA	NIEA	NRW	SEPA
Acts (primary legislation)					
Environmental Protection Act 1990	✓		✓	✓	✓
Waste and Contaminated Land (Northern Ireland) Order 1997			✓		
Regulations (secondary legislation)					
Controlled Waste (England and Wales) Regulations 2012	✓	✓		✓	
Controlled Waste (Registration of Carriers and Seizure of Vehicles) Regulations 1991		✓	✓		✓
Waste Batteries and Accumulators Regulations 2009	✓		✓	✓	✓
Waste Regulations (Northern Ireland) 2011		✓	✓		
Waste (England and Wales) Regulations 2011	✓	✓		✓	
Waste (Scotland) Regulations 2012		✓			✓
Hazardous Waste					
Hazardous Waste (Amendment No. 2) Regulations (Northern Ireland) 2015		✓	✓		
Hazardous Waste (England and Wales) Regulations 2005	✓	✓		✓	
Special Waste Regulations 1996		✓			✓
Landfill					
Landfill Regulations (Northern Ireland) 2003		✓	✓		
Landfill (England and Wales) Regulations 2002	✓	✓		✓	
Landfill (Scotland) Regulations 2003		✓			✓
Permits and Licences					
Environmental Permitting (England and Wales) Regulations 2016	✓	✓		✓	
Waste Management Licencing (Amendment) Regulations (Northern Ireland) 2022		✓	✓		
Waste Management Licencing (Scotland) Regulations 2011		✓			✓
Guidance					
HSE publications and guidance					
Net Regs Environmental guidance					
SEPA guidance notes					
Waste Resource Action Programme (WRAP) hosted by CIRIA					
Organisations with information and guidance on their websites	✓	✓	✓	✓	✓

*Key

CIRIA	Construction Industry Research and Information Association
EA	Environment Agency
HSE	Health and Safety Executive
LA	Local Authorities
NIEA	Northern Ireland Environment Agency
NRW	Natural Resources Wales
SEPA	Scottish Environment Protection Agency

Overview

The most recent data from 2018 for non-hazardous and construction and demolition waste from the Department for the Environment, Farming and Rural Affairs (Defra) established that the UK produced 137.8 million tonnes of construction, demolition and excavation (CD&E) waste (in England, this was 119.4 million tonnes).

This chapter gives a general overview of the legal framework for the regulation of waste. It outlines how waste is defined and classified for the purpose of disposal and introduces the waste hierarchy. This chapter also gives you an overview of the permits that are required for the management of waste and the documentation required under the duty of care for its safe and environmentally-sound disposal. It provides an overview of various types of waste, including hazardous waste, waste electrical and electronic equipment (WEEE) and waste batteries.

10.1 Introduction

Of the over 100 million tonnes of waste produced by the CD&E sector per year, a proportion ends up in landfill sites. This waste has further environmental impacts after its disposal (such as the generation of greenhouse gases) that contribute to climate change.

Waste is a by-product of the inefficient use of valuable resources that could be reused on site or recycled. The simple operation of cutting a brick in half, or sawing the end off a piece of wood, is producing waste if that half brick or off-cut is not reused. Waste costs money to produce, in terms of the materials thrown away, and also money to dispose of, along with costs for the skip or lorry to remove it. Landfill tax rates will continue to rise to provide a financial incentive to reduce the amount of waste being sent to landfill.

Research carried out on housing projects has indicated that the true cost of disposing of the material is around 10 times the cost of the skip. This includes the labour to fill the skip and the cost of purchasing the materials in the first place. Typically skips account for 0.5% of the build cost. Overall waste and its disposal can amount to around 5% of the project costs. This is valuable lost profit for the project. Making the best use of materials and resources is therefore essential in reducing waste costs.

The incorrect or inappropriate disposal of waste (such as fly-tipping) is illegal, unsightly and can damage the environment for many years. To tackle the problem of waste crime, Local Authorities (LAs) have the power to stop and search vehicles thought to be involved in illegal waste activity. The Environmental Improvement Plan (EIP), Goal 5, sets out targets and commitments to maximise resources and minimise waste, as follows.

- Eliminate avoidable waste by 2050 and double resource productivity by 2050.
- Explore options for the near elimination of biodegradable municipal waste to landfill from 2028.
- Eliminate avoidable plastic waste by 2042.
- Seek to eliminate waste crime by 2042.
- To halve 'residual' waste (excluding major mineral waste) produced per person by 2042 (waste that is sent to landfill, put through incineration or used in energy recovery in the UK, or that is sent overseas to be used in energy recovery).

The residual waste target is underpinned by the following interim targets, to be achieved by 31st January 2028:

- Reduce residual waste (excluding major mineral waste) produced per person by 24%, and in total tonnes by 21%.
- Reduce municipal residual waste per person by 29%.
- Reduce residual municipal food waste per person by 50%.
- Reduce residual municipal plastic waste per person by 45%.
- Reduce residual municipal paper and card waste per person by 26%.
- Reduce residual municipal metal waste per person by 42%.
- Reduce residual municipal glass waste per person by 48%.

Insufficient attention to the generation of waste at design stage, during procurement and poor control and supervision by site management, including improper or unsafe systems of work on site, increase the production of waste. Waste materials lying around on a construction site are a major safety hazard, increasing the fire risk and the potential for trips and slips, which can have a huge impact on both people's work and personal lives. Legislation imposes conditions and obligations on how contractors in the construction industry must plan, manage and dispose of any waste produced during work both on and off site.

10.1.1 Waste regulation authorities

The principal waste regulation authorities are the Environment Agency (EA) in England, the Northern Ireland Environment Agency (NIEA), the Scottish Environment Protection Agency (SEPA), and Natural Resources Wales (NRW). The agencies took over these functions from LAs, and their responsibilities include dealing with the application and enforcement of waste management licences, permits and exemptions, waste carriers' licences and the duty of care regime *(refer to 10.9)*. LAs are also responsible for dealing with local air pollution control (LAPC) matters related to waste (such as the issue of permits for crushing equipment).

The improper disposal of waste is illegal, and can lead to prosecution and even imprisonment. The maximum penalty for waste crime is a £50,000 fine and up to 12 months' imprisonment in the magistrates' court, or five years' imprisonment and an unlimited fine at the crown court.

If someone finds your waste flytipped, and you cannot prove that you have not dumped it and that you have complied with all the necessary requirements, you could be fined up to £50,000 and face up to five years in prison. Failure to produce waste transfer notes can result in a fixed penalty notice of £300.

10.2 Important points

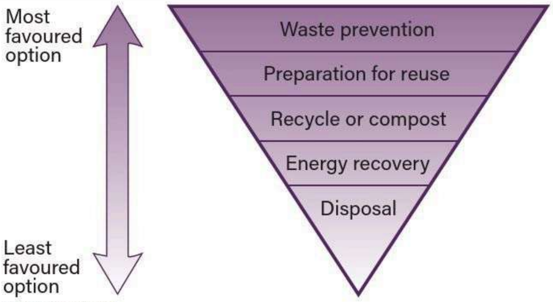
- Producers of waste must correctly identify whether surplus materials are waste and classify it as non-hazardous or hazardous using a waste classification code, also referred to as LoW (List of Waste) or EWC (European Waste Catalogue) code.
- With reference to the standard industry classification (SIC) code, producers of waste must correctly identify the SIC and which area of construction the waste originates from (for example, construction of buildings, civil engineering or specialist construction activities).
- Producers of waste have a legal duty of care to ensure that it is passed on to an authorised person who has the correct technical competence and holds a relevant environmental permit or licence.
- All contractors who carry or collect construction and demolition waste must have a waste carrier's licence.
- All waste transfers must be supported by the correct document (a controlled waste transfer note) for non-hazardous waste. The transfer of hazardous waste requires a consignment note. Both of these documents must include a declaration that the producer of the waste has considered the waste hierarchy in deciding to dispose of the material.
- In Wales, producers of 500 kg or more of hazardous waste (such as oils or asbestos) must register their premises with the NRW. Failure to do so could result in a fine of up to £5,000.
- All waste treatment or disposal facilities should have an environmental permit (England and Wales) or waste management licence (Northern Ireland and Scotland) unless they have a registered exemption from the EA, NRW, NIEA or SEPA.
- Where materials are treated or processed on site before being suitable for putting back into the works, consideration must be given as to whether this activity requires an environmental permit or registered exemption. Compliance with schemes, including the Contaminated Land: Application In Real Environments (CL:AIRE) Code of Practice or the waste and resources action programme (WRAP) quality protocol for the production of aggregates from inert waste, could avoid the requirement of an environmental permit for reuse of the processed material.

10.3 Waste hierarchy

Article 4 of the revised Waste Framework Directive (WFD) (2008/98/EC) requires that all reasonable measures should be taken to prevent waste and consider the waste hierarchy when waste is transferred.

The waste hierarchy is a series of steps for dealing with waste in order of priority and signifies the relative environmental benefits that can be made at each stage.

The waste hierarchy shows that the highest priority is waste prevention or reduction so that the need for other options (such as reuse, recycling and energy recovery) would be dramatically reduced.



Waste hierarchy

The Waste (England and Wales) Regulations now require that you declare on the waste transfer documentation that you have considered the waste hierarchy in the management of the waste.

Examples of applying the waste hierarchy on a construction project for waste reduction could include the following.

- Prevention or minimisation.
  - Designing the project to suit standard product sizes to avoid site cutting.
  - Designing the project's landscape levels to avoid excavation materials going off site.
  - Specification to allow the use of recycled materials.
  - Pre-assembling components off site or using pre-cast sections.
  - Not over-ordering materials.
  - Reducing the amount of packaging.
  - Ordering materials at the size required, to avoid off-cuts.
  - Promoting employee awareness of environmental matters.
  - Requiring sub-contractors to have a waste management policy.
  - Not over-excavating.

10

- Reuse.
  - Reusing soil for landscaping.
  - Using off-cuts of timber for alternative uses.
  - Using brick rubble as hardcore.
  - Investigating local environmental or construction work where other materials might be reused.
- Recycling.
  - Crushing waste concrete to use as hardcore.
  - Recycling asphalt planings as road sub-base or temporary surfacing.
  - Recycling scrap metal, glass and waste oil.
  - Segregating waste materials to a separate well-planned area.
  - Recycling timber by donation for educational use or social enterprise.
  - Recycling office waste (such as paper, cans and plastics).
- Recovery.
  - Sending waste for composting or for energy recovery (such as timber off site to be shredded for use as biomass fuel, or composting).
- Disposal.
  - Sending canteen or office waste for disposal to a local landfill site.

Office recycling

Plasterboard recycling

Segregated waste

Segregated waste streams

The recycling or treatment of waste on or off site is likely to require a waste management permit (England and Wales) or licence (Northern Ireland and Scotland), or registered exemption *(for further details on waste management permitting refer to 10.12).*

Social enterprise for waste – National Community Wood Recycling Project

The National Community Wood Recycling Project (NCWRP) was founded in 2003 to help set up and develop a nationwide network of wood recycling social enterprises.

The National Builders Collection Scheme was set up to market the service to building companies in 2010, allowing the NCWRP to become self-funding and offer support to enterprises without the need for grants or fundraising.

There are currently 30 enterprises operating across the country, collectively forming Community Wood recycling, and collecting around 20,000 tonnes of wood every year.

The NCWRP website provides details of locations where wood can be recycled.

10.4 Site waste management plans

SWMPs on construction projects used to be a legal requirement under the Waste Management Plan Regulations. However, in line with the Government's Red Tape Challenge, designed to remove unnecessary legislation to free up businesses, **the regulations were repealed**.

The aim of a SWMP was to reduce the amount of waste produced on construction sites, by setting out how building materials and any resulting waste was managed during a project. The plan was updated during the construction stages, which recorded and confirmed how materials were reused, recycled or disposed of. The introduction of the regulations had aimed to:

1. improve resource efficiency and reduce waste
2. prevent fly-tipping.

SWMPs are often now referred to as resource management plans (RMPs). Whilst there is no legal requirement for a SWMP, implementing one helps manage materials more effectively and helps to reduce material waste and cost savings. Regardless of whether a SWMP (or RMP) is used, all construction companies have a duty of care towards managing their waste under section 34 of the Environmental Protection Act 1990. A SWMP/RMP can raise the profile of waste planning, improve the environmental awareness of the workforce, and ensure compliance with the regulations.

10.5 The Environment Targets (Residual Waste) (England) Regulations 2023

The government published new legislation, the Environmental Targets (Residual Waste) (England) Regulations, which came into force on 30th January 2023 to address the residual waste volumes needing to be halved by 2042 in a target based on 2019 numbers.

The regulations, which extend to England and Wales but apply to England, come under the Environment Act 2021, which requires the setting of deliverable waste targets to help government improve the state of the environment on land and at sea. This includes having less waste alongside a more sustainable use of resources. These regulations are expected to have significant implications for waste policy and the development of recycling and residual waste facilities in years to come.

The waste target is for the reduction of residual waste (excluding major mineral wastes), on a kilogramme per capita basis, by 50% by 2042 from 2019 levels. This will be measured as a reduction from the 2019 level, which was updated to 574 kg per capita following updated evidence received by Defra following consultation. The residual waste long-term target is that by the end of 31st December 2042 the total mass of residual waste for 2042 does not exceed 287 kg per head of population in England.

10.6 Defining waste

Waste is defined in the Environmental Protection Act as: any substance which constitutes a scrap material, an effluent or other unwanted surplus arising from the application of any process or any substance or article, which requires to be disposed of, which has been broken, worn out, contaminated or otherwise spoiled.

In practice, however, this definition has been tested by case law; determining whether or not a substance is waste depends on applying the right legal tests. A substance can be classed as waste even if the producer still has a use for it or if other people are prepared to pay for it. This is important because whether or not a material is waste determines whether a complex body of legal rules and restrictions govern what can be done with it.

Surplus materials are generally not waste while they remain as the original manufactured product and do not need to be re-processed. However, construction or demolition waste that has been generated as part of the works could be classified as waste until it has been processed (for example, crushed and screened) and recovered back into the permanent works. In these cases there will be various requirements to demonstrate that the material has achieved an **end of waste** status.

Materials that meet the end of waste test must satisfy the following criteria.

- The material must be converted into a distinct and marketable product.
- It can be used in the same way as an ordinary soil or aggregate.
- It can be used without creating any negative environmental effects.

Aggregates manufactured from construction and demolition waste complying with the WRAP quality protocol, for example, can normally demonstrate that the material has achieved an end of waste status. Likewise, complying with the CL:AIRE Development Industry Definition of Waste Code of Practice (DoW CoP) will also help you to achieve end of waste status for the treatment and use of contaminated excavated materials *(for more on the CL:AIRE DoW CoP, refer to 10.13 and to Chapter E08 Resource efficiency, 8.5.2.)*

10.7 Describing and classifying waste

Wastes will always fall into one of four categories.

- Always non-hazardous, known as **absolute non-hazardous** (for example, clean bricks or glass).
- Always hazardous known as **absolute hazardous** (for example, insulating materials containing asbestos).
- May, or may not, be hazardous and need to be assessed known as **mirror hazardous** and **mirror non-hazardous** (for example, contaminated soils).

What makes a waste hazardous is whether it contains any hazardous substances above certain thresholds that make it display a certain hazardous property. 15 different hazardous properties exist, from HP1 to HP15, together with persistent organic pollutant (POP) criteria (for example, HP1 is explosive and HP6 is acute toxicity). The threshold concentration levels for the relevant hazardous properties are defined within the guidance on the classification and assessment of waste (WM3). Waste can be referred to as **active**, for hazardous waste, or **inactive**, for inert waste (such as rocks and bricks).

For further information refer to the joint agency technical guidance *Waste classification – Guidance on the classification and assessment of waste* (WM3).

Where there is doubt over the presence or levels of hazardous materials the waste must first be classified as both mirror hazardous and mirror non-hazardous. Further testing then has to take place to confirm the presence and levels of hazardous content to ensure the final classification is accurate.

The most appropriate method of classifying waste, where it needs to be assessed, is to identify the hazardous constituents or chemicals in the waste, determine the hazard statement codes and hazardous properties of these substances and then to use their concentrations to identify whether they exceed the threshold levels of any of the hazardous properties. The safety documentation supplied with any product should provide sufficient information to make this assessment. For contaminated soils, however, detailed testing would need to be carried out by competent staff. The waste classification code, also referred to as **LoW (List of Waste)** or **EWC (European Waste Catalogue)** codes, are classification codes for common types of waste. Arranged in 20 sections, Section 17 contains the codes for construction and demolition waste.

Appendix A includes the full Section 17 list of wastes codes relating to construction and demolition waste.

The list of wastes refers to hazardous and non-hazardous entries. Where an entry is marked with an asterisk it is classified as **hazardous waste** if it meets one of the following criteria.

- The hazardous entry makes no reference to hazardous substances (for example, 17 06 05* construction materials containing asbestos). These types of entries are always hazardous and are called **absolute hazardous entries (AH)**.
- The hazardous entry refers to a waste containing hazardous substances where the concentration levels of these hazardous substances exceed the threshold limits (for example, 17 05 03* soil and stones containing hazardous substances). These entries are called **mirror hazardous entries (MH)** as it depends on the concentration of hazardous substances to determine whether they are hazardous or not. Wastes containing hazardous substances below the threshold limits are non-hazardous.
- The lists of wastes also contain **mirror non-hazardous entries (MN)**. The mirror non-hazardous usually (but not always) has a defined link to its mirror using the words 'other than those mentioned in....' (for example, 07-01-12 sludges from on-site effluent treatment other than those mentioned in 07-01-11 MN).

e.g. Typical construction wastes from Section 17 of the list of wastes

- Bricks – 17 01 02 Non-hazardous.
- Concrete – 17 01 01 Non-hazardous.
- Wood – 17 02 01 Non-hazardous.
- Plasterboard – 17 08 02 Non-hazardous (but must be segregated from other wastes).
- Insulation containing asbestos – 17 06 01* Mirror hazardous (will be hazardous if concentrations exceed thresholds).
- Contaminated soil – 17 05 03* Mirror hazardous (will be hazardous if concentrations exceed thresholds).

Other types of construction wastes not covered in Section 17

- Waste hydraulic oil – 13 01 13* Absolute hazardous.
- Mixed canteen waste – 20 03 01 Non-hazardous.

The disposal of hazardous waste arising from construction operations, or from contaminated land, is dealt with under the Hazardous Waste (England and Wales) Regulations 2005, the Hazardous Waste (Amendment No. 2) Regulations 2015 or the Special Waste Regulations 1996 (in Scotland) *(for further information refer to 10.11).*

All waste transfer documentation must include the relevant EWC six-digit code that describes and categorises the waste, and the relevant SIC code describing the area of construction from where the waste originated.

For the purpose of disposal of **waste to landfill** there are three classes of waste.

Inert waste that will not decompose to produce greenhouse gases (such as rubble, concrete and glass).

Non-hazardous waste that will rot and decompose (such as timber, food and paper) and does not contain hazardous substances.

Hazardous waste that has substances in sufficient concentration to make it possess one or more of the 15 hazardous properties (such as explosive (HP1) or acute toxicity (HP6)) and is hazardous to human health or the environment (such as asbestos or oil).

There are strict criteria for the acceptance of waste at each of these three types of landfill site. Certain types of waste, particularly contaminated soil, would have to be tested in order to demonstrate that it meets the relevant waste acceptance criteria. The Landfill Directive introduces a hierarchy of waste characterisation and testing known as the **waste acceptance procedures**. The levels are shown below.

Level 1. Basic characterisation. A thorough determination, according to standardised analysis and behaviour-testing methods, of the leaching behaviour and/or characteristic properties of the waste.

Level 2. Compliance testing. A periodic testing of regularly arising wastes by simpler standardised analysis and behaviour-testing methods to determine whether a waste complies with permit conditions, and whether a waste with known properties has changed significantly. The tests focus on the main variables and behaviour identified by basic characterisation.

Level 3. On-site verification. This constitutes checking methods to confirm that a waste is the same as that which has been subjected to compliance testing and that which is described in the accompanying documents. It may merely consist of a visual inspection of a load of waste before and after unloading at the landfill site.

Before sending waste to landfill, waste producers and landfill operators must ensure that they know all of the properties of the waste, relevant to its potential for pollution or harm to health, and the options for the management of the waste.

There are certain types of waste that are banned from disposal to landfill (shown below). These must either be recovered, recycled or disposed of in other ways (for example, incineration).

- Any liquid waste.
- Infectious medical or veterinary waste.
- Whole or shredded used tyres.
- Waste that might cause a problem in the landfill (such as hot or chemically active waste).
- Any waste that does not meet the waste acceptance criteria for that class of landfill.

The term **difficult waste** has come into common use, and applies to wastes that require handling in a particular way. It is not used in legislation, but is used to cover a grey area of wastes. Although not classed as hazardous by reference to the Hazardous Waste Regulations, this type of waste is nevertheless difficult to dispose of. Examples of difficult waste are shown below.

- **Invasive plants** that are waste materials, both soil and plant matter (contaminated, for example, with Japanese knotweed or giant hogweed), which can only be disposed of at sites that are specifically licensed to receive them.
- **Contaminated soil** that is a mixture of soils, stones, rubble and polluting substances, which could be hazardous depending on thresholds, and could be a range of things left over from former use of the site.
- **Gypsum and plasterboard wastes** that, when mixed with biodegradable waste, can produce hydrogen sulphide gas in landfill, which is both toxic and odorous.

Gypsum and plasterboard waste

The land filling of gypsum and other high sulphate-bearing wastes, together with biodegradable waste, has been prohibited. All gypsum waste should be segregated from biodegradable waste before being sent to landfill. It should be noted that plasterboard waste itself (unless contaminated) is not hazardous waste but must be segregated.

10.8 Treatment of waste

The Environmental Permitting (England and Wales) Regulations, Landfill Regulations (Northern Ireland) 2003 and the Landfill Regulations (Scotland) 2003 state that non-hazardous waste must now be treated before being sent to landfill. In practical terms, *treatment* is applying the waste hierarchy to reduce the quantity of waste that ends up in landfill. Treatment must satisfy all three criteria of a three point test, as shown in the table on the right.

1.	Be a physical, thermal, chemical, or biological process (including sorting).
2.	Change the characteristics of the waste.
3.	Be carried out in order to: – reduce the volume of the waste – reduce the hazardous nature of the waste – facilitate handling of the waste – enhance recovery of the waste.

On construction sites, in practical terms, this can be achieved by setting up segregated skips and separating out **(sorting)** any wastes that can be reused or recycled, which will change the **characteristics of the original waste stream**. This in turn will aid in **reducing the volume** of the waste, **facilitating the handling** of the waste and **enhancing recovery** of the waste destined for landfill.

Hazardous wastes are required to be stored and disposed of separately from non-hazardous wastes. Sending your waste to a transfer station or recycling facility, for sorting and recovery prior to the residual waste being sent to landfill, will also satisfy these treatment requirements.

Excavated materials that are to be treated on or off site are generally considered to be waste and the treatment facility operator must have an appropriate environmental permit or register a waste exemption allowing that particular treatment of the excavated materials *(for further explanation of this issue refer to 10.11, 10.12 in this chapter, and to 8.5.2 and 8.5.3 in Chapter E08 Resource efficiency).*

Persistent organic pollutants (POPs) are poisonous chemical substances that break down slowly and get into food chains as a result.

Hexabromocyclododecane (HBCD) has now been listed as a persistent organic pollutant (POP).

Waste containing POPs should be destroyed at end of life.

For polystyrene blocks used in buildings that use the flame retardant HBCD, the best way of ensuring destruction is by incineration.

In England, municipal waste incinerators are likely to be sufficient to destroy HBCD.

10.8.1 Waste (Scotland) Regulations 2012

These Scottish Government regulations expand the duty of care requirements, over and above those required in England, Northern Ireland and Wales, and require all waste producers, including construction companies, to present dry recyclable material for separate collection. To ensure waste producers can comply with this requirement, the waste industry in Scotland has to provide services that enable the separate collection of dry recyclables such as glass, metals, plastics, paper and cardboard.

Across the UK, food businesses, in urban areas, producing large amounts of food waste (> 50 kg per week) are required to present that food waste for separate collection. Construction canteens come under the definition of a food business.

Segregated waste streams

In Scotland and Northern Ireland the regulation has been amended and this requirement now extends to smaller food businesses that produce between 5 kg and 50 kg per week. Only businesses producing very small amounts of food waste (< 5 kg per week) are exempt from the duty altogether. There is an expectation that this new requirement will be expanded to cover the whole of the UK in the near future.

The use of macerators to dispose of food waste in the sewer system has been banned from 1 January 2016, except for domestic premises and food producers in rural areas.

10.9 Duty of care and waste carrier registration

If you are involved in managing waste you have a legal duty of care. The duty of care applies to everyone involved in handling the waste, from the person who produces it to the person who finally recovers or disposes of it.

You have a legal responsibility to ensure that you produce, store, transport and dispose of your business waste without harming the environment. Duty of care is one of the main ways to avoid environmental harm and combat fly-tipping and means that:

- waste has to be stored in a secure location and measures taken to prevent its escape
- waste has to be passed to an authorised person holding a valid licence or permit
- any waste passed to an authorised person must be supported by the relevant documentation.

10.9.1 Registration of waste carriers, brokers and dealers (England and Wales)

The Waste Regulations (England and Wales) implement a system for the registration of:

- waste carriers (those who move waste)
- waste brokers (those who arrange the movement and/or disposal on behalf of others)
- waste dealers (those who use an agent to buy and sell waste).

It is possible for a single registration to cover all of this work.

There are two classes of registration, known as lower tier and upper tier.

10.9.1.1 Lower tier

Those registered in the lower tier are known as **specified persons**. This group includes all those who are currently registered as professionally exempt.

In general this refers to waste authorities, charities, voluntary organisations and those who only manage wastes from agricultural premises, animal by-product wastes or wastes from mines or quarries.

Lower tier registration is not an option for construction and demolition companies that move their own waste. The guidance states that: anyone who normally and regularly carries their own business waste (excluding construction and demolition waste) will need to register in the upper tier. There is currently no fee for lower tier registration; the registration will be valid until it is revoked or cancelled.

10.9.1.2 Upper tier

If you are a construction or demolition company and you move your own waste you must register in the upper tier. There will be a fee payable for upper tier registration. Upper tier registration will be valid for three years unless it is revoked or cancelled. If you are a waste carrier, broker or dealer but you are not a specified person, you will need to register in the upper tier.

10.9.2 Registration of waste carriers (Northern Ireland)

A similar system of registration, with slight differences, applies in Northern Ireland under the Controlled Waste (Registration of Carriers and Seizure of Vehicles) Regulations 1991.

www **An application for registration as a carrier, broker or dealer can be made on the Environment Agency, Natural Resources Wales or Northern Ireland Environment Agency websites or by downloading the appropriate form.**

10.9.3 Registration of waste carriers (Scotland)

The Controlled Waste (Registration of Carriers and Seizure of Vehicles) Regulations 1991 in Scotland require that if you transport waste within the UK, in the course of your business or in any other way for profit, you must register as a carrier of waste with the local waste regulation authority. You must register even if you only carry your own company's waste or carry waste on an infrequent basis. This applies whether you are a self-employed contractor, part of a partnership or a company.

Registration only needs to be made in the area where your company has its head office. All other offices will be covered by this one registration. Application for registration as a waste carrier must be made on the prescribed form, which is obtainable from the local waste regulation authority office. If you have applied for registration but have not yet received the documentation, you will be deemed to be registered and may carry waste.

The law demands that waste carriers keep a copy of their registration document on their vehicle. Do not accept photocopies of registration documents as proof of registration. If you have any doubt as to whether the carrier is registered or not, ask the appropriate waste regulation authority.

10.10 Controlled waste and transfer notes

All waste subject to the provisions of the Environmental Protection Act is known as controlled waste and includes waste from domestic, commercial and industrial premises as well as hazardous waste. Under the duty of care waste must be passed to an authorised person (a holder of a waste carrier or waste management permit or licence).

When non-hazardous waste is transferred to an authorised person this must be supported by a waste transfer note. This also applies even if you have produced and carry waste yourself. The transfer of hazardous waste must be supported by a consignment note *(refer to 10.11)*.

Refer to Appendix B for a sample of a controlled waste transfer note.

Whenever you pass non-hazardous waste on to someone else, you will have to declare on the waste transfer note that you have applied all reasonable measures to apply the waste management hierarchy. The duty of care waste transfer note must:

- include a declaration that you have taken all measures to apply the waste management hierarchy
- include the appropriate SIC code, identifying the area of construction that generated the waste
- give a description of the waste, including the six-digit EWC code
- state the quantity
- state how the waste is contained, whether loose or in a container and, if in a container, the kind of container
- identify the carrier of the waste
- state the date, time and location of transfer
- identify the disposal site
- contain your signature and the signature of the authorised person receiving your waste.

A copy of the transfer note must be kept for a minimum of two years. A copy must also be given to the disposal site representative, who will sign the documents to say where, when and how the waste will or has been disposed of.

If you use a registered carrier to dispose of your waste for you, the transfer note must contain all the points described above and, in addition, state the following.

- Name and address of the carrier, their licence registration number and issuing authority.
- Place of transfer.

If you use a registered carrier to remove your waste, they will raise and distribute the necessary documentation and will:

- give you a copy to keep
- keep a copy for themselves
- deliver your waste and a copy of the document to the management at the disposal site.

You must keep all controlled, non-hazardous waste transfer documentation (such as waste transfer notes and consignment notes) and records accessible for two years.

10.11 Hazardous waste and consignment notes

Sites producing or storing less than 500 kg per year must comply with the Hazardous Waste (England and Wales) Regulations 2005. There is no requirement to register your site in England, Northern Ireland or Scotland.

www **For further information visit the Government legislation website.**

The Hazardous Waste (England and Wales) Regulations in England, the Hazardous Waste (Amendment No. 2) Regulations (Northern Ireland 2015, and the Special Waste Regulations 1996 in Scotland, require that all hazardous waste must be segregated from non-hazardous waste.

The following controls should be adopted to comply with the regulations.

- Different types of hazardous wastes should be segregated, to identify the quantities and types on the hazardous waste consignment note.
- Mixing of different types of hazardous waste should be avoided as this may inadvertently create an explosive or fire risk, particularly in warm weather.
- The mixing of hazardous waste with non-hazardous waste to dilute the material below the threshold concentration is banned.
- Packaging or containers contaminated with hazardous substances should be treated as hazardous waste unless it can be shown that the concentration (including the packaging) is below the threshold limits.

Segregated hazardous waste

Where individual products are combined to form a substance (such as adhesives and resins) then each component should be considered for its hazardous properties and disposed of accordingly. Resins are often inert when set, so leaving materials to dry before disposal may make them non-hazardous.

10.11.1 Removal of hazardous waste

When hazardous waste is removed from site, a document called a **hazardous waste consignment note** (special waste consignment note in Scotland) must be prepared. The consignment note must stay with the hazardous waste until it reaches its final destination, and must be filled in first by the producer of the waste. The remaining parts will be completed depending on your role in the waste process (see below table).

In Scotland, a special waste consignment note must be obtained from the local office of the Scottish Environment Protection Agency (SEPA).

Your role	Part(s) you must complete
Producer	A and B
Holder (stores the waste)	A and B
Carrier (collects and transports the waste)	C
Consignor (hands the waste to the carrier)	D
Consignee (receives the waste for recycling or disposal)	E

The hazardous waste consignment note must be completed in the following way. The producer must complete parts A and B, including:

- the consignment note code
- the address the waste is being taken to
- the address the waste is being removed from
- the details of who produced the waste
- the process that gave rise to the waste, including the SIC code
- waste details, including:

– description of the waste
– LoW or EWC code
– quantity
– physical form
– chemical components
– hazard code
– container type
– UN identification number.

The carrier must check the waste and journey details entered by the producer, and complete section C, including:

- vehicle and carrier details
- multiple collection details, if relevant.

The consignor must check sections A, B and C, and complete D, in the form of a consignor's certificate. They must keep one copy, pass one to the producer or holder, and hand the other back to the carrier. The carrier must then promptly deliver the waste to the consignee. On arrival, the carrier gives the consignee two copies, and they must then check and either accept or reject the waste. After checking the waste and the note to ensure it's correct and complete, the consignee completes part E, and enters:

- the EWC code
- whether the EWC code is accepted or rejected
- the quantity of each code received
- the waste management operation details.

**The consignment note coding system for England and Wales was changed on 1 April 2016. The consignment note code is a unique code for each consignment of hazardous waste and is always in the same format of the first six letters or numbers of the company name, followed by a forward slash, and then five further letters or numbers (for example CJTILE/A0001).*

10.11.2 Coding format for hazardous waste consignment notes

The part that applies to you depends on your role in the waste process, as follows in the below table. A consignment note code must be created (in the format XXXXXX/YYYYY) and entered onto the consignment note. The table shows the coding format for hazardous waste consignment notes in England and Wales.

Coding format for hazardous waste consignment notes		
Code	**Explanation**	**Example**
XXXXXX/	First six letters and/or numbers (not symbols or spaces) of the name of the company entered in Part A2 of the consignment note. This must be followed by a forward slash '/'. ***Note:*** *if your company name has less than six letters and/or numbers you must assign the letter 'Q' to the remaining characters.*	CJTILE/A0001
YYYYY	Exactly five digits made up of numbers and/or letters (not symbols or spaces) of your choice. ***Note:*** *each code can only be used once from the address in Part A2. You must change the 'YYYYY' to create a different code each time waste is consigned from that premises.*	
Additional letters		
An additional letter must be assigned at the end of the consignment note code for certain types of consignment: 'V' to waste removed from ships 'XXXXXX/YYYYYV' 'F' to fly-tipped waste 'XXXXXX/YYYYYF' 'D' to waste moving under a consignee return derogation 'XXXXXX/YYYYYD' 'P' to continuous piped waste 'XXXXXX/YYYYYP' Additional letters must also be assigned to rejected loads. The consignment note form contains two extra grey boxes for these additional letters.		

In Northern Ireland and Scotland, a copy of completed consignment notes must be received by NIEA or SEPA at least 72 hours before the waste is due to leave site.

Completed consignment notes are valid for 28 days after the anticipated date of collection.

You must keep records of all hazardous waste documentation in a register for three years after the waste is transferred.

For an example of a hazardous waste consignment note refer to Appendix C.

10.12 Environmental permits and exemptions

10.12.1 Environmental permitting in England and Wales

The Environmental Permitting Regulations (England and Wales) (EPR) combine the system of waste management licensing previously regulated under the Waste Management Licensing Regulations, and the system of permitting installations in the Pollution Prevention and Control Regulations (England and Wales). The EPR also now include the provision of permits to deal with groundwater protection, water discharges and flood defence activities.

For further information on environmental permits visit the Government website.

For further information on water protection refer to Chapter E07 Water management and pollution control.

The EPR specify which waste activities require an environmental permit and allow some waste operations to be exempt from requiring a permit: certain operations covered by other legislation are excluded from these permitting arrangements. An environmental permit is required for seven different classes of **regulated facility**, five of which relate to waste (the other two cover groundwater and water discharges).

Installations. Generally these are facilities at which industrial, waste and intensive farming, falling (mainly) under the Integrated Pollution Prevention and Control Directive, are carried out. These include landfill sites, asphalt plants and concrete batching plants.

Waste operation. Any other waste activity that is not defined as an installation will be classed as a waste operation. This includes the depositing, treatment or recycling of waste that is not exempt under the Environmental Permitting Regulations (such as waste transfer stations) including the treatment of contaminated land.

Mobile plant. Mobile equipment carrying out an activity listed in Schedule 1 of the EPR or a waste operation (such as crushers).

Mining waste operation. The management of mining extraction waste that may include the mining waste facility.

Radioactive substances. The keeping and management of radioactive material (including radioactive apparatus) or the storage and disposal of radioactive waste.

It is an offence to operate a regulated facility without the relevant permit, or breach permit conditions. You could be fined up to £50,000 and imprisoned for up to five years for this offence.

Two types of permit can be applied for.

Standard permits for certain standard types of waste operation (such as a waste transfer station or mobile treatment plant for the treatment of waste soils). Each type of standard permit has standard rules and risk criteria that have to be met to comply with the permit.

Bespoke permits for more complex operations, which are specifically relevant to the waste facility or operation but cannot meet the criteria required in a standard permit.

If you can't meet the requirements of a standard permit then you may need to apply for a bespoke permit.

Exemptions are for works considered to be too trivial or below the threshold limit for a permit. Many need to be registered *(refer to 10.12.1.3)*.

10.12.1.1 Environmental permit applications

The starting point is to understand what type of waste activity will be carried out and who regulates it, and then to complete an application to the appropriate regulator. Part 2 of Schedule 1 to the EPR lists regulated facilities that are installations and mobile plant. They include Part A(1) work regulated by the EA (England) or NRW (Wales) and Part A(2) and Part B work regulated by LAs. The EA or NRW generally regulate higher-risk work that can pollute more than one media (such as water and air), whereas LAs regulate work that contributes to air pollution. The table below defines which authority is responsible for issuing permits for regulated facilities associated with construction.

Type of regulated facility	Regulator and where to send permit applications
Installations	
Landfill sites	EA or NRW
Asphalt plant	EA or NRW
Concrete batching plants	LA
Waste operation	
Waste transfer stations	EA or NRW
Use of waste in construction	EA or NRW
Mobile plant	
Mobile plant for the treatment of contaminated soils	EA or NRW
Mobile plant for crushing and screening demolition waste	LA

Mobile plant for the treatment of waste soils and contaminated material is a waste operation regulated under a standard permit SR2008 No. 27 by the EA or NRW. In addition to holding the permit the operator is required to prepare a site-specific deployment form, which sets out in detail the type of technology used and specified work at the site. The treatment of contaminated soil and/or contaminated waters requires a mobile treatment permit (MTP).

A MTP is used to regulate a mobile plant activity that involves treatment either in situ or off site. The permit sets out the type and extent of work that can be carried out. A site-based permit has to be used where a mobile plant permit is not applicable. The environmental permit can be either a standard rules permit or a bespoke permit, depending upon the type of treatment and site location.

Operators who want to treat contaminated soil and/or contaminated waters using their mobile plant permit at a particular site must submit a site-specific deployment application (the deployment form and supporting information). The deployment application details site-specific information and potential impacts arising from the proposed use of the mobile plant. The operator must demonstrate that the activity will not cause pollution of the environment, harm to human health or serious detriment to local amenities.

Following treatment under a MTP the material would normally cease to be waste, providing it was excavated and treated on the site where it will be used or is part of a remediation cluster. Where this is not the case then an environmental permit or registered exemption would be required.

For flowcharts providing further guidance on permitting the use or reuse of soils or aggregates refer to Appendix D.

The regulator has four months from receipt of the permit application and all supporting information to make a determination. You have the right of appeal to the Secretary of State should your permit not be granted, or the permit is granted but you are not happy with the conditions that have been imposed. Pre-application discussions with the regulator can help in improving the quality of permit applications so early contact with them is advisable. The Waste (England and Wales) Regulations cover a new permit condition to require waste to be managed in accordance with the waste hierarchy.

Only a person who is in control of a regulated facility may obtain or hold an environmental permit. This person is called the operator. To obtain an environmental permit you may have to prove that you have the appropriate technical competence *(refer to 10.12.1.2)* to be able to carry out the relevant work and fulfil the obligations of an operator.

The regulator will consider the following.

- Whether your management systems are adequate.
- If your site is run by someone who is technically competent.
- Any convictions that you, or other persons in your business, may have for pollution offences.
- If you have taken steps to meet the possible costs of the duties of the permit.

10.12.1.2 Technical competence

Certain types of waste management activity require the operator to demonstrate technical competence. They are called relevant waste operations. These include, for example, landfill sites, transfer stations and certain work involving hazardous waste.

You will be able to show that you are a technically competent operator if you can satisfy one of the following.

- Compliance with an approved industry scheme. There are currently two approved schemes for operators of relevant waste operations, the:
  1. Chartered Institution of Wastes Management (CIWM) or Waste Management Industry Training and Advisory Board (WAMITAB) scheme
  2. Environmentally Sensitive Areas/EU scheme.
- Holding an appropriate certificate of technical competence (CoTC) from the WAMITAB, which issues a range of certificates for the managers of most types of waste management site. This can be checked on the WAMITAB CoTC database.
- You have previously completed an environmental assessment for non-CoTC work.

10.12.1.3 Exemptions from environmental permitting in England and Wales

There are a number of exemptions from environmental permitting for certain waste activities that are not seen as a threat to the environment. The general requirements and descriptions of these exemptions are set out in Schedule 3 of the Environmental Permitting Regulations.

The Environmental Permitting Regulations have significantly changed the descriptions and references for exemptions, which are now grouped into four categories.

Use of waste. This includes the recovery or reuse of waste for a purpose. These exemptions will have a 'U' reference (for example, *U1 – Use of waste in construction*).

Treatment of waste. This includes the treatment of material for the purpose of recovery. These exemptions will have a 'T' reference (for example, *T5 – Screening and blending of waste for the purposes of producing an aggregate or soil and associated prior treatment*). Note that some wastes produced under this exemption may then be used under another exemption (such as the production of aggregates) that is then used under a *U1 – Use of waste in construction exemption.*

Disposal of waste. This includes the disposal of certain types of waste onto land. These exemptions will have a 'D' reference (for example, *D1 – Deposit of waste from dredging of inland waters*).

Storage of waste. This includes the storage of waste at a location other than where it was produced pending its recovery or disposal. These exemptions will have an 'S' reference (for example, *S2 – Storage of waste in a secure place*).

Refer to Appendix E for materials and quantity thresholds for U1 or T5 exemptions.

These exemptions generally exclude hazardous waste and there will be conditions on most exemptions granted. Some examples of the conditions included in the exemption are shown below.

- Limits on the quantities and time periods for the temporary storage of waste produced on site for reuse on that site.
- Limits on the quantities for the spreading, on land, of waste soil.
- Certain other precautions that must be taken.

You may deal with waste under an exemption subject to the conditions imposed, but you must make sure that you do not pollute the environment or cause harm to anyone's health. To obtain a waste management exemption under Schedule 3 of the Environmental Permitting Regulations, an application must be made, together with the appropriate fee, to the correct waste regulation authority. In most cases this is the EA or NRW.

However, if you want to crush bricks, tiles and concrete you may need to register a *T7 – Treatment of waste bricks, tiles and concrete by crushing, grinding or reducing in size* exemption with your Local Authority. Also note that the equipment will need to have a valid mobile plant permit, which must be obtained from the Local Authority in which the business is situated. The information required varies considerably depending on the relevant exemption applied for. Notifications for more complex exemptions will generally require a form to be completed and must be supported by the following type of information.

- Name and address of applicant for the exemption.
- Location of the site where the waste activity is being carried out.
- Details of relevant planning permissions.
- Description, type and analysis of the waste.
- Intended use for the waste.
- The relevant fee.

A number of exemptions also have time constraints and cannot be renewed before the expiry date. For example, under a *U1 – Use of waste in construction* exemption, you cannot register more than once at any one place during the three-year period from first registration.

For flowcharts providing further guidance on exemptions from permitting for the use or reuse of soils or aggregates refer to Appendix D.

10.12.2 Waste management licensing in Northern Ireland and Scotland

In Northern Ireland the waste licensing system is created under Part II of the Waste and Contaminated Land (Northern Ireland) Order 1997 and controlled through the Waste Management Licensing (Amendment) Regulations (Northern Ireland) 2022. In Scotland, the existing waste management licensing system is controlled under Section 33 of Part II of the Environmental Protection Act and the Waste Management Licensing (Scotland) Regulations 2011.

A waste management licence is required if you deposit, recover, treat or dispose of controlled waste. If you do any of these things without a licence, or a licence exemption, you could be fined and/or sent to prison. Obtaining a full waste management licence can be a lengthy and costly exercise. To obtain a waste management licence you have to prove that you are a fit and proper person. Applications for waste management licences should be made to NIEA or SEPA, who will consider the following.

- Any convictions that you, or other persons in your business, may have for pollution offences.
- If your site is run by someone who is technically competent.
- If you have taken steps to meet the possible costs of the duties of the licence.

10.12.2.1 Technical competence

In Northern Ireland the certificate of technical competence (CoTC) remains the primary means for managers of appropriate waste management facilities to demonstrate technical competence, in accordance with the Waste Management Licensing Amendment) Regulations (Northern Ireland). In Scotland, under the Waste Management Licensing (Scotland) Regulations there is no longer a legal requirement for a technically-competent person to hold a CoTC. These were previously regulated by the WAMITAB under Waste Management Licensing Regulations. CoTCs remain an appropriate qualification to demonstrate competence in Scotland and can be used on a voluntary basis. Operators in Scotland should contact their SEPA regulatory officer in the first instance to check other competence arrangements.

www **For further details of competence requirements visit the WAMITAB website.**

10.12.2.2 Waste management licence applications

An application for a waste management licence can only be made if planning permission has been granted. To obtain a waste management licence, you should apply to the waste licensing office of NIEA or SEPA and ask the following questions.

Questions for NIEA or SEPA

- How do you apply for the licence?
- What information do they require?
- How can you show that you are a fit and proper person, including technical competence?
- How much is the application fee for the type of site you wish to run?

NIEA and SEPA have four months from when all information was received to consider the application. You have the right of appeal to the Secretary of State should your licence not be granted, or the licence is granted but you are not happy with the conditions imposed.

10.12.2.3 Exemptions from waste management licensing in Northern Ireland and Scotland

There are a number of waste management exemptions set out in Schedule 2 of the Waste Management Licencing (Amendment) Regulations (Northern Ireland) and Schedule 1 of the Waste Management Licencing (Scotland) Regulations. These exemptions generally exclude hazardous waste and there will be conditions (examples below) on most exemptions granted.

- Limits on the quantities and time periods for the temporary storage of waste produced on site for reuse on that site.
- Limits on the quantities for the spreading, on land, of waste soil.
- Certain other precautions that must be taken.

You may deal with waste under an exemption subject to the conditions imposed, but you must make sure that you do not pollute the environment or cause harm to anyone's health.

To obtain a waste management exemption, an application must be made, together with the appropriate fee, if appropriate, to NIEA or SEPA. The information required varies considerably depending on the relevant exemption being applied for. NIEA and SEPA make a distinction between simple and complex exemptions and the notification process is different for each type. Notifications for complex exemptions will generally require a form to be completed and must be supported by the following type of information.

- Name and address of applicant for the exemption.
- Location of the site where the waste activity is being carried out.
- Details of relevant planning permissions.
- Description, type and analysis of the waste.
- Intended use for the waste.
- The relevant fee.

A number of the exemptions are also required to be renewed on an annual basis. Further advice should be sought from NIEA or SEPA.

10.13 Waste avoidance protocols

10.13.1 CL:AIRE Development industry Code of Practice (CoP)

The CL:AIRE (Contaminated Land: Applications in Real Environments) Code of Practice (CoP) has implications for waste management of contaminated land, but is essentially intended to avoid waste creation through treatments that convert materials for alternative uses to reduce landfill.

For detailed information on the CL:AIRE CoP refer to Chapter E08 Resource efficiency.

10.13.2 WRAP quality protocol for the production of aggregates from inert waste

This protocol provides a formalised quality control procedure for the production of aggregates from recovered inert waste.

For detailed information on the recycled aggregates quality protocol refer to Chapter E08 Resource efficiency.

10.14 Asbestos waste

There are slight differences in the respective legislation for England and Wales, Northern Ireland and Scotland. Contractors who work in any of these areas must comply with the relevant national legislation.

In essence, there is a duty of care placed on everyone in the waste disposal chain. Clients, whether they are commercial or domestic, have a responsibility to ensure that the waste is handled by competent and, as appropriate, registered contractors and is disposed of correctly. You can be prosecuted if waste is fly-tipped.

Where asbestos has been removed as part of a project, a prudent contractor would ensure that proof of disposal formed part of the information that was given to the client. In England, where a site waste management plan or demolition audit is utilised it should include details of the disposal of the asbestos waste and the waste carrier.

The principal contractor should ensure that disposal records are passed to the principal designer to ensure that this information is placed in the health and safety file to confirm that the asbestos has been transported and disposed of legally.

10.14.1 Classification

Asbestos waste is almost always hazardous waste (special waste in Scotland). There are two main scenarios to consider.

1. A single type of waste that has been separated.
2. A mixed waste.

For a single type of waste that has been separated, the hazardous waste concentration limit for asbestos in a waste is 0.1%, or more asbestos (weight for weight). All asbestos-containing materials (ACMs) used in building materials will contain asbestos at or above this concentration, and will be classed as hazardous waste.

For mixed waste, where more than one type of waste is present, mixing is prohibited. Dilution caused by mixing cannot be used to justify a non-hazardous classification. The waste containing the asbestos that is mixed with other waste should be assessed separately and would retain its hazardous status.

Examples of hazardous waste include the following.

- Another waste, such as soil, crushed secondary aggregate, or mixed construction or demolition waste containing one or more piece(s) of ACM, of a size visible to the naked eye.
- The mixing of a soil from a contaminated area of a site, containing 0.1% or more asbestos fibres, with other soil from the same site in a stockpile; on a gravimetric (weight for weight) basis.

Note: *asbestos waste also includes contaminated building materials and tools that cannot be decontaminated; personal protective equipment; and damp rags used for cleaning. If in doubt, always treat waste as 'hazardous' or 'special'.*

The below table shows the levels of asbestos, and what level is required to class it as hazardous across the UK nations.

England and Wales	• Asbestos waste is 'hazardous waste' when it contains more than 0.1% asbestos. • The Hazardous Waste (England and Wales) Regulations (HWR) apply. • You must complete a hazardous waste consignment note. • Contact the EA for more information in England, and the NRW in Wales.
Scotland	• Asbestos waste is 'special waste' when it contains more than 0.1% asbestos. • The Special Waste Amendment (Scotland) Regulations 2004 (SWAR) apply. • Complete a hazardous waste consignment note. • Contact the SEPA for more information.
England, Scotland and Wales	• All asbestos waste is subject to Schedule 2 of the Control of Asbestos Regulations 2012, and most waste is subject to the Carriage of Dangerous Goods and Use of Transportable Pressure Equipment Regulations 2009 (CDG). • CDG does not apply to firmly-bound asbestos, asbestos cement, or articles with asbestos reinforcement which do not release hazardous or respirable fibres easily. However, HWR and SWAR still apply. CDG applies for all other asbestos waste.

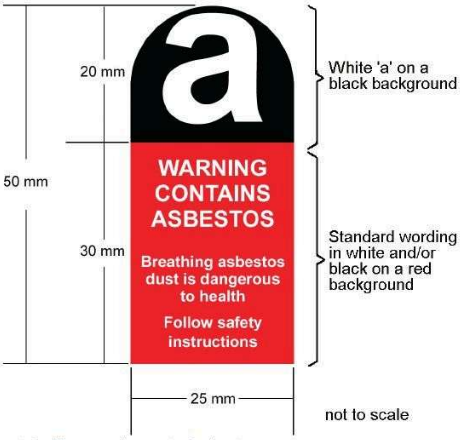
www **For further guidance on the classification and assessment of waste visit the Government website.**

For further information on asbestos refer to Chapter B09 Asbestos.

10.14.2 Management on the site of production

A refurbishment and demolition asbestos survey is required when any premises, or part of it, needs refurbishment or demolition. Where asbestos waste has been identified, details of the type, estimated quantities and disposal options should be included in the project site waste management plan (or alternative document if such a plan is not in place).

Asbestos cement sheets or other asbestos cement products should be double bagged in UN-approved packaging, with a CDG hazard label and asbestos code information visible, or double-wrapped in 1,000-gauge polythene sheeting and labelled accordingly if they are being taken directly from the place of production to landfill (for example, not being taken to and stored temporarily at a contractor's premises or a transfer station). Alternatively, waste can be placed in a lockable skip, but it must be easily cleanable: it is unacceptable to throw sheeting over a standard skip.



Labelling requirement of asbestos waste

Where a skip is used on site it must not be used for consolidating asbestos waste from other sites, as this would introduce more complex hazardous waste requirements. The site would then be recognised as a hazardous waste transfer station requiring an environmental permit.

Skips containing asbestos should be lockable and secured at all times. It is considered good practice to double bag or wrap waste from internal work to ensure that no asbestos is dropped when the waste is carried outside. If a van is being used to carry small volumes of asbestos waste, it must be double bagged and the bags secured (for example, by putting them in a bin with a lid). This must only be transported in a segregated compartment just for asbestos, and the vehicle must be easily cleanable.

Any fibrous asbestos removed from an enclosure must be double bagged. Bagged waste should consist of an inner red bag and an outer clear bag. These should be correctly marked. The packaging (bagging) and labelling must satisfy the requirements of current legislation and, where asbestos waste is to be transported by road, there is further legislation controlling its safe movement (International Carriage of Dangerous Goods by Road Regulations (ADR)). Containers should be designed and constructed so as to retain the asbestos without any spillage or loss during normal handling.

Where the asbestos removal contractor uses labels on waste or removed asbestos, the label must be clearly and indelibly printed so that the words in the lower half can be easily read. These words must be printed as specified in the diagram above. The label must be firmly affixed to the packaging using an adhesive type label, or directly printed onto the packaging (plastic bag).

10.14.3 Removal from the site of production

Contractors must ensure the following for hazardous waste.

- It is consigned from the place where the waste was produced, using a hazardous waste consignment note.
- The waste must be transported by a registered waste carrier.
- In Wales, the hazardous waste producer must also register, with NRW, the place where the hazardous waste was produced (unless they are exempt from registration). This requirement does not apply to England or Scotland.
- The waste is taken directly from the site of production to a suitably authorised facility within 24 hours. The facility should hold either a relevant environmental permit or an exemption from a permit.
- In Scotland and Northern Ireland the 72 hour pre-notification will apply.

10.14.4 Suitable authorised facilities

Any place where asbestos waste is taken must hold an environmental permit that authorises the acceptance of asbestos waste. This will normally be one of the following.

- A landfill with a separate cell set aside for asbestos.
- A temporary storage facility (operated by the asbestos contractor, or a waste contractor) that stores waste before transfer to landfill.

An asbestos contractor that takes asbestos waste back to their own premises must also hold an environmental permit. Although there is an exemption for storage at a temporary collection point controlled by the producer, this is restricted to bonded asbestos only, and excludes businesses providing a waste management service to customers (such as those contracted to remove asbestos).

- Asbestos waste must not be put into bays or transferred between different skips or containers.
- Keep all handling to a minimum (either manual or mechanical). Loading shovels must not be used to move asbestos waste.
- Keep designated skips secure and locked when asbestos waste is not being deposited into them.
- Operators must have emergency procedures to deal with delivered, non-conforming asbestos waste. This could include waste that is not properly contained in bags or skips (for example, unwrapped asbestos sheets in open skips). Sites should also have the capability to deal with appropriate non-conforming waste and spillages (for example, broken bags). Records should be kept of non-conforming waste.

- Do not use a skip that has been used to store asbestos for non-asbestos use, unless it has been verified that the skip has been fully decontaminated.

Any place that receives hazardous waste, including exempt facilities in England and Wales, is a consignee and required to send quarterly returns to the EA or NRW.

10.14.5 Crushed secondary aggregate

Companies that produce crushed secondary aggregate from construction or demolition waste are not authorised to accept and process waste containing asbestos. They should have effective waste acceptance and screening procedures to prevent this from happening. The secondary aggregates they produce should not contain asbestos.

A secondary aggregate containing the following is likely to remain hazardous waste, and will be subject to waste controls, which would prevent its use in construction:

- one or more piece of ACM (as detailed in Classification of waste), or
- any concentration of asbestos resulting from the processing of pieces of ACM or a waste containing asbestos fibres at a hazardous concentration.

10.15 Waste electrical and electronic equipment (WEEE)

The Waste Electrical and Electronic Equipment Regulations 2013 (WEEE Regulations) apply to England, Northern Ireland, Scotland and Wales, and designates safe and responsible collection, recycling and recovery procedures for all types of electrical and electronic waste (whether whole or broken) that is destined for disposal. This is widely known as WEEE or e-waste.

There are 10 broad categories of WEEE currently covered in the regulations, and the scope now extends to further categories of electric and electronic equipment (EEE).

www **To discover the 10 broad categories currently covered within the WEEE Regulations, refer to Schedules 1 and 2; for a detailed list of the EEE, refer to Schedules 3 and 4.**

A producer selling electrical equipment for non-household use (such as equipment used by a construction company) is obliged to finance the collection, treatment and recycling in an environmentally sound manner of:

- any waste electrical equipment replaced (with equivalent or similar function) by the electrical equipment sold, if it was originally purchased before 13 August 2005, whether supplied by this or another producer
- the electrical equipment the producer sold on or after 13 August 2005 when it is eventually discarded as WEEE.

The collection, treatment and recovery may be undertaken either by the producer, or by their producer compliance scheme (PCS), which they must register with. The PCS should be registered with the appropriate waste regulation authority (EA, NRW, NIEA or SEPA).

Equipment distributors (retailers, wholesalers, mail order or internet dealers) do not have any specific obligations for non-household electrical equipment. However, the PCS registration information should be passed on so that the end user (such as a construction company) can properly dispose of the item at the end of its life.

Every year an estimated two million tonnes of WEEE items are discarded by householders and companies in the UK. This includes most products that have a plug, or need a battery.

10.15.1 End-user responsibilities for WEEE

A construction company's role in the WEEE Regulations means that, to dispose of WEEE, the following actions must be taken.

- It must be segregated from other types of waste for disposal.
- If the waste electrical equipment was purchased before 13 August 2005, and is being replaced with new equivalent equipment, then ask the producer for details of its PCS and collection arrangements.
- If the waste equipment is not being replaced with new equivalent equipment, or the PCS cannot be traced, then you must pay to transfer the waste equipment to an approved authorised treatment facility that can accept waste electrical equipment (such as a licensed transfer station).
- Any waste transferred to an authorised collector or waste carrier must meet all of the normal requirements for duty of care (such as waste carrier's licence, transfer notes and licensed treatment facilities, for example waste transfer station) approved by the waste regulation authority.

Waste electrical equipment

10

10.16 Waste batteries

The construction industry is a large user of batteries in many types of vehicles, plant and equipment. Currently the recycling rate for portable batteries is around 44%. The Waste Batteries and Accumulators Regulations 2009 apply to the UK and Northern Ireland. They set out requirements for waste battery collection, treatment, recycling and disposal for all types of battery, and aim to reduce the impact of waste batteries on the environment and increase recycling. The regulations define three types of batteries (listed below) and by chemistry as lead-acid, nickel-cadmium or other.

Portable batteries that are sealed, can be hand carried and are neither an automotive battery or accumulator nor an industrial battery. Examples of a portable battery include AA or AAA-type batteries or the battery used to power a laptop, mobile phone, calculator and car remote-locking or alarm controls. The UK currently uses around 30,000 tonnes of portable batteries each year, which come in various sizes and chemistries.

Industrial batteries used for industrial or professional purposes, such as the battery used as a source of power and propulsion to drive the motor in an electric forklift, batteries used to power hybrid motor vehicles, and batteries in solar-powered road signs. These batteries comes in a wide range of chemistries, and are subject to producer take-back.

Waste portable batteries

Automotive batteries used for starting or the ignition of a vehicle engine, or for powering the lights of a vehicle. Due to the high levels of lead in these batteries, there has traditionally been a high recycling rate, currently of around 95%.

10.16.1 Take-back of waste batteries

These regulations require that distributors of **portable batteries** (for example, retail stores) have a duty to take back waste portable batteries through facilities such as in-store waste-battery bins. This requirement does not apply to distributors who supply less than 32 kg of batteries per year. A producer of **industrial batteries** is obliged to provide for the take-back of waste industrial batteries free of charge from the end user in the following circumstances.

- If the producer has supplied new industrial batteries to that end user.
- Where, for any reason, the end user is not able to return waste industrial batteries to the supplier who supplied the batteries, providing the waste batteries are the same chemistry as the batteries the producer places on the market.
- If the end user is not purchasing new batteries, and a battery with the same chemistry as the one being returned has not been placed on the market for a number of years, then the end user's entitlement is to be able to contact any producer to request take-back.

10.16.2 Guidance for battery users

An important aim of the regulations is to enable end users of industrial batteries to have them treated and recycled at no cost to themselves. As with any waste, the final holders must comply with the duty of care for waste by ensuring that all waste transfers are passed to authorised persons, together with the correct waste transfer documentation. Waste should only be carried by a licensed waste carrier.

www **For further information on the Waste Batteries and Accumulators Regulations visit the Government legislation website.**

10.16.3 Lithium-ion batteries

Lithium-ion (Li-ion) batteries come in a variety of shapes, sizes and chemistries. There are two basic categories: non-rechargeable batteries containing lithium metal, and rechargeable batteries containing salts of lithium. They are used for electric vehicles, but many smaller batteries are used in everyday devices, such as mobile phones, vaping devices, laptops, digital cameras, electric scooters and e-bikes.

While these batteries are designed to align with sustainability, they pose some serious safety risks. They are highly flammable, and if damaged or not fully sealed, or if they come into contact with water, they can quickly catch fire. Li-ion batteries should never be incinerated due to the risk of explosion.


Explosion of a Lithium-ion battery installed in a mobile Bluetooth speaker

They can be recycled, but it is essential that this is done safely, and they must go to a specialist recycling centre. The fire risk is particularly dangerous during waste transportation or the recycling process. If a fire occurs in a vehicle or at the facility, it could cause extensive damage, and even be life-threatening. Li-ion batteries need to be disposed of at a hazardous waste collection site: if disposed of in general waste or a recycling container, they could easily come into contact with water. This could happen if it is raining, if a crushed bottle of liquid leaks, or if condensation forms within the container.

Appendix A – European Waste Catalogue six-digit list of wastes codes

Section 17 – Construction and demolition waste

A six-figure list of wastes code for the type of waste being removed MUST be written on every waste transfer note (for example, skip/muck away tickets).

17 01 Concrete, bricks, tiles and ceramics	
17 01 01	Concrete.
17 01 02	Bricks.
17 01 03	Tiles and ceramics.
17 01 06*	Mixtures of, or separate fractions of concrete, bricks, tiles and ceramics containing hazardous substances.
17 01 07	Mixtures of concrete, bricks, tiles and ceramics other than those mentioned in 17 01 06.
17 02 Wood, glass and plastic	
17 02 01	Wood.
17 02 02	Glass.
17 02 03	Plastic.
17 02 04*	Glass, plastic and wood containing or contaminated with hazardous substances.
17 03 Bituminous mixtures, coal tar and tarred products	
17 03 01*	Bituminous mixtures containing coal tar.
17 03 02	Bituminous mixtures other than those mentioned in 17 03 01.
17 03 03*	Coal tar and tarred products.
17 04 Metals (including their alloys)	
17 04 01	Copper, bronze, brass.
17 04 02	Aluminium.
17 04 03	Lead.
17 04 04	Zinc.
17 04 05	Iron and steel.
17 04 06	Tin.
17 04 07	Mixed metals.
17 04 09*	Metal waste contaminated with hazardous substances.
17 04 10*	Cables containing oil, coal tar and other hazardous substances.
17 04 11	Cables other than those mentioned in 17 04 10.
17 05 Soil (including excavated soil from contaminated sites), stones and dredging spoil	
17 05 03*	Soil and stones containing hazardous substances.
17 05 04	Soil and stones other than those mentioned in 17 05 03.
17 05 05*	Dredging spoil containing hazardous substances.
17 05 06	Dredging spoil other than those mentioned in 17 05 05.
17 05 07*	Track ballast containing hazardous substances.
17 05 08	Track ballast other than those mentioned in 17 05 07.
17 06 Insulation materials and asbestos-containing construction materials	
17 06 01*	Insulation materials containing asbestos.
17 06 03*	Other insulation materials consisting of or containing hazardous substances.
17 06 04	Insulation materials other than those mentioned in 17 06 01 and 17 06 03.
17 06 05*	Construction materials containing asbestos.
17 08 Gypsum-based construction material	
17 08 01*	Gypsum-based construction materials contaminated with hazardous substances.
17 08 02	Gypsum-based construction materials other than those mentioned in 17 08 01.
17 09 Other construction and demolition wastes	
17 09 01*	Construction and demolition wastes containing mercury.
17 09 02*	Construction and demolition wastes containing polychlorinated biphenyls (PCBs) (for example, PCB-containing sealants, PCB-containing resin-based floorings and PCB-containing sealed glazing units).
17 09 03*	Other construction and demolition wastes (including mixed wastes) containing hazardous substances.
17 09 04	Mixed construction and demolition wastes other than those mentioned in 17 09 01.

Entries marked with * are either potentially hazardous (hazardous mirror entry) and will require testing to determine if hazardous properties are present or definitely hazardous (absolute) and must be disposed of as hazardous waste.

Appendix B – Example of a controlled waste transfer note

Duty of care: waste transfer note

Keep this page and copy it for future use. Please write as clearly as possible.

Section A – Description of waste

A1 Description of the waste being transferred

List of Waste Regulations code(s)

A2 How is the waste contained?

Loose ☐ Sacks ☐ Skip ☐ Drum ☐

Other ☐

A3 How much waste? For example, number of sacks, weight

Section B – Current holder of the waste – Transferor

By signing in Section D below I confirm that I have fulfilled my duty to apply the waste hierarchy as required by Regulation 12 of the Waste (England and Wales) Regulations 2011 Yes ☐

B1 Full name

Company name and address

Postcode SIC code (2007)

B2 Name of your unitary authority or council

B3 Are you:

The producer of the waste? ☐

The importer of the waste? ☐

The local authority? ☐

The holder of an environmental permit? ☐

Permit number

Issued by

Registered waste exemption? ☐

Details, including registration number

A registered waste carrier, broker or dealer? ☐

Registration number

Details (are you a carrier, broker or dealer?)

Section C – Person collecting the waste – Transferee

C1 Full name

Company name and address

Postcode

C2 Are you:

The local authority? ☐

C3 Are you:

The holder of an environmental permit? ☐

Permit number

Issued by

Registered waste exemption? ☐

Details, including registration number

A registered waste carrier, broker or dealer? ☐

Registration number

Details (are you a carrier, broker or dealer?)

Section D – The transfer

D1 Address of transfer or collection point

Postcode

Date of transfer (DD/MM/YYYY)

D2 Broker or dealer who arranged this transfer (if applicable)

Postcode

Registration number

Time(s)

Transferor's signature

Name

Representing

Transferee's signature

Name

Representing

You must keep all controlled, non-hazardous waste transfer documentation for two years.

Appendix C – Example of a hazardous waste consignment note

Form HWCN01v112

The Hazardous Waste Regulations 2005: Consignment Note

Environment Agency

PRODUCER'S/HOLDER'S/CONSIGNOR'S COPY (Delete as appropriate)

PART A Notification details

1 Consignment note code: /

2 The waste described below is to be removed from (name, address, postcode, telephone, e-mail, facsimile):

3 The waste will be taken to (name, address and postcode):

4 The waste producer was (if different from 2) (name, address, postcode, telephone, e-mail, facsimile):

PART B Description of the waste If continuation sheet used, tick here

1 The process giving rise to the waste(s) was:

2 SIC (2007) for the process giving rise to the waste: . /

3 WASTE DETAILS (where more than one waste type is collected all of the information given below must be completed for each EWC identified)

Description of waste	List of wastes (EWC code)(6 digits)	Quantity (kg)	The chemical/biological components in the waste and their concentrations are:		Physical form (gas, liquid, solid, powder, sludge or mixed)	Hazard code(s)	Container type, number and size
			Component	Concentration (% or mg/kg)			

The information given below is to be completed for each EWC identified

EWC code	UN identification number(s)	Proper shipping name(s)	UN class(es)	Packing group(s)	Special handling requirements

PART C Carrier's certificate

(If more than one carrier is used, please attach schedule for subsequent carriers. If schedule of carriers is attached tick here. ☐)

I certify that I today collected the consignment and that the details in A2, A3 and B3 are correct and I have been advised of any specific handling requirements.

Where this note comprises part of a multiple collection the round number and collection number are: /

1 Carrier name:

On behalf of (name, address, postcode, telephone, e-mail, facsimile):

2 Carrier registration no./reason for exemption:

3 Vehicle registration no. (or mode of transport, if not road):

Signature

Date D D M M Y Y Y Y Time H H M M

PART D Consignor's certificate

I certify that the information in A, B and C has been completed and is correct, that the carrier is registered or exempt and was advised of the appropriate precautionary measures. All of the waste is packaged and labelled correctly and the carrier has been advised of any special handling requirements.

I confirm that I have fulfilled my duty to apply the waste hierarchy as required by Regulation 12 of the Waste (England and Wales) Regulations 2011.

1 Consignor name:

On behalf of (name, address, postcode, telephone, e-mail, facsimile):

Signature

Date D D M M Y Y Y Y Time H H M M

PART E Consignee's certificate (where more than one waste type is collected all of the information given below must be completed for each EWC)

Individual EWC code(s) received	Quantity of each EWC code received (kg)	EWC code accepted/rejected	Waste management operation (R or D code)

1 I received this waste at the address given in A3 on: Date D D M M Y Y Y Y Time H H M M

2 Vehicle registration no. (or mode of transport if not road):

3 Where waste is rejected please provide details:

I certify that waste permit/exempt waste operation number:

authorises the management of the waste described in B at the address given in A3.

Where the consignment forms part of a multiple collection, as identified in Part C, I certify that the total number of consignments forming the collection are:

Name:

On behalf of (name, address, postcode, telephone, e-mail, facsimile):

Signature

Date D D M M Y Y Y Y Time H H M M

You must keep all hazardous waste documentation for three years.

Appendix D – Waste flowcharts for the reuse of construction materials (soils and aggregates)

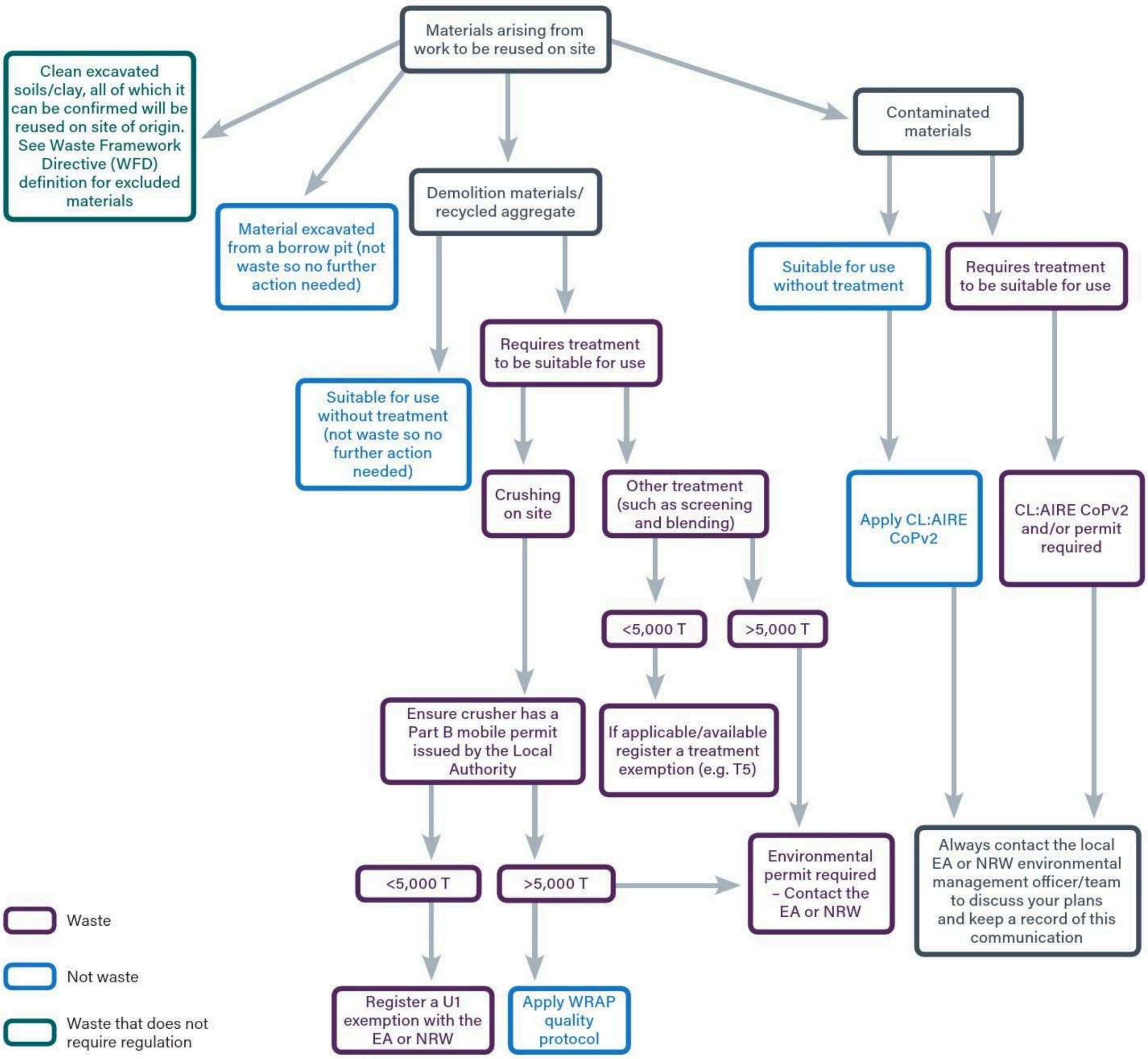
The following three flowcharts will help you to decide whether use and reuse of soil and aggregate materials in construction works is a waste activity or not and what you need to do to ensure legal compliance whilst minimising the regulatory burden. These flowcharts may also assist you in completion of a site waste management plan (SWMP) when assessing reuse or recycling options during design and construction phases. These flowcharts are applicable in England and Wales.

These flowcharts are not intended to cover exhaustive criteria of waste; however they do cover the typical scenarios that are commonly faced on construction and civil engineering projects.

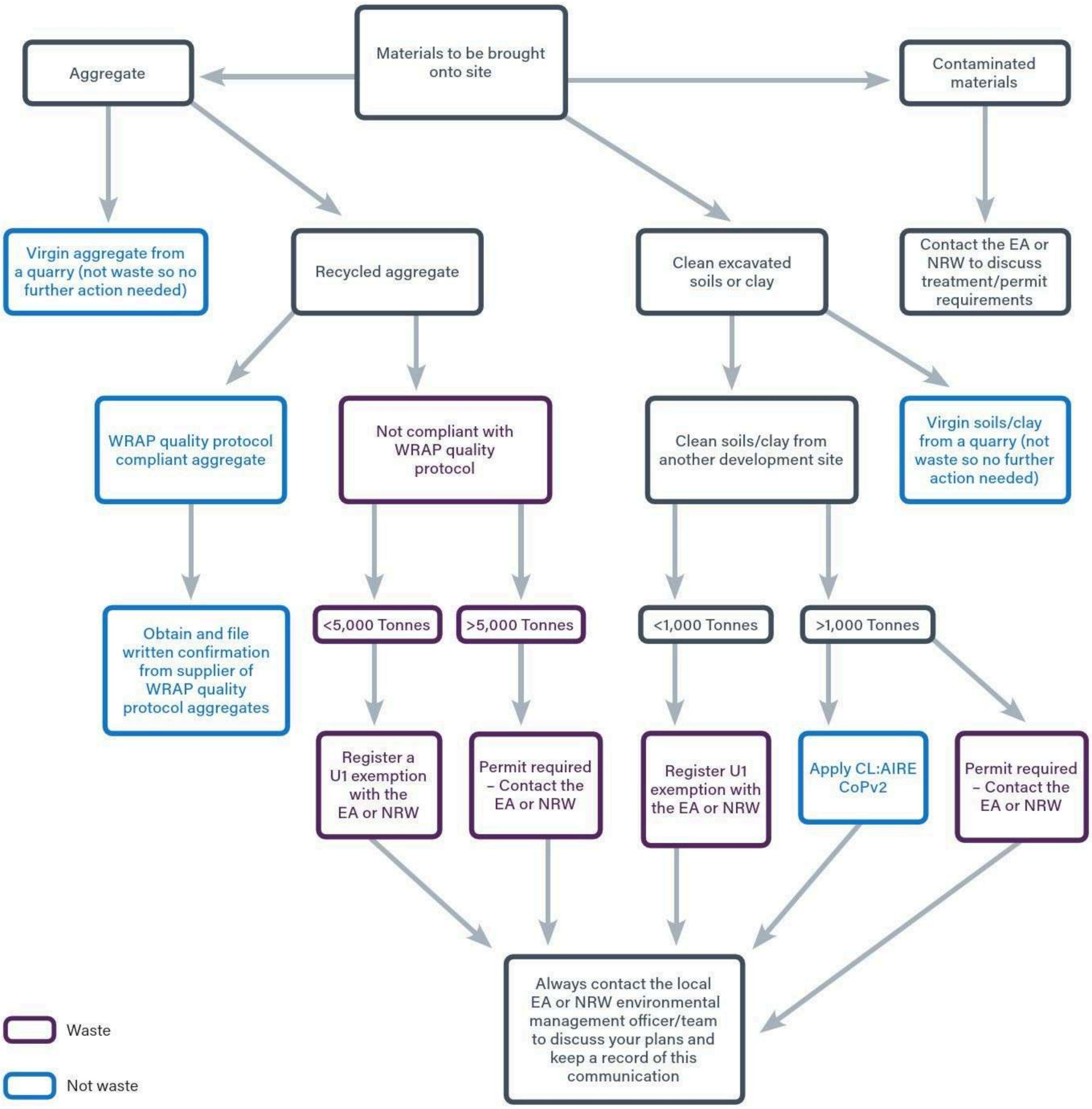
For simplicity, these flowcharts have been split up into three priority operations (*see below and on the following pages*).

EA refers to the Environment Agency in England, and NRW refers to Natural Resources Wales in Wales.

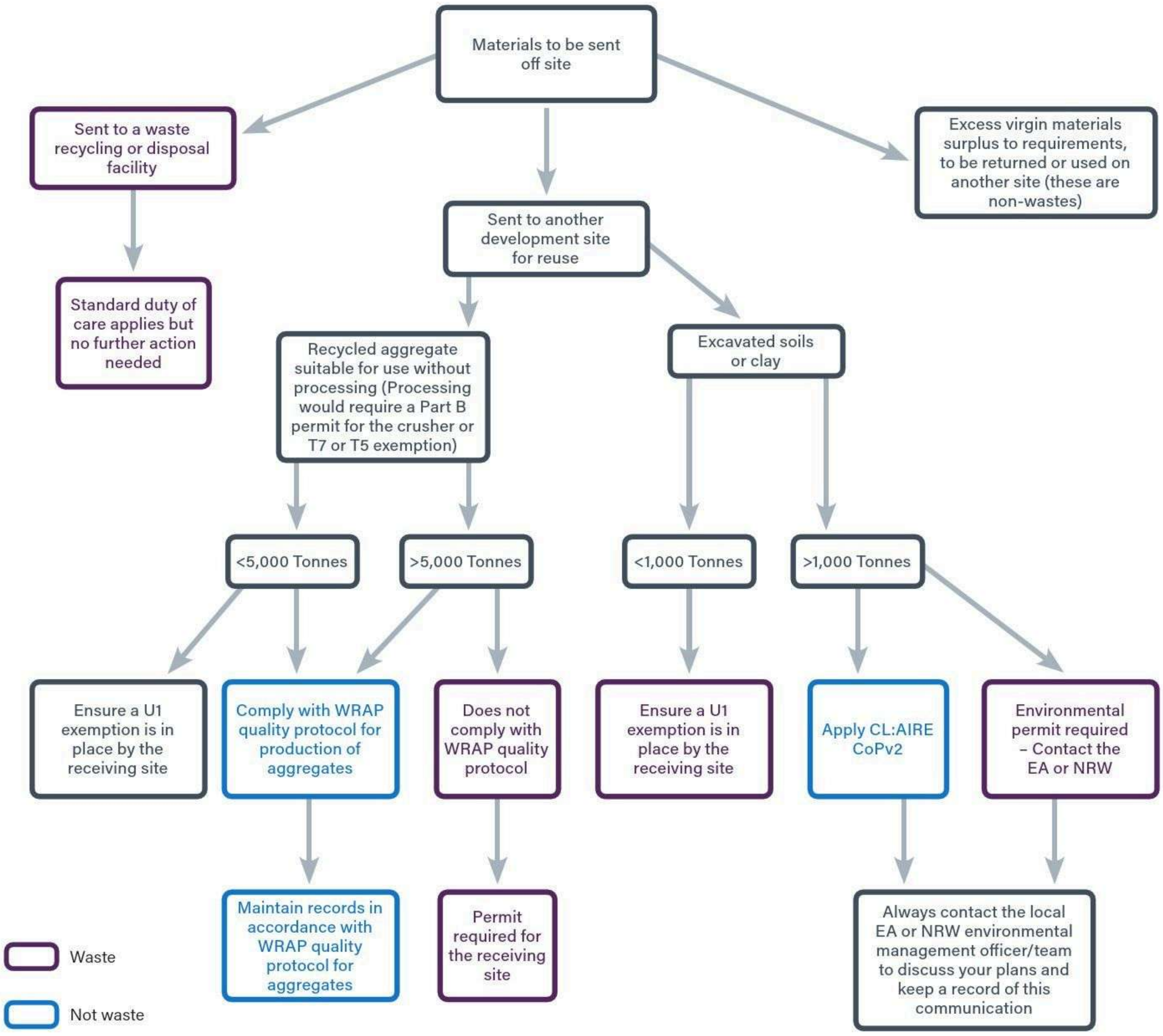
Materials arising from work to be reused on site



Materials to be brought onto site



Materials to be sent off site



Appendix E – Waste exemption materials and thresholds

Exemption U1 – Use of waste in construction

This exemption is referenced in the Environmental Permitting (England and Wales) Regulations, and allows the use of suitable wastes for small-scale construction instead of using virgin raw materials.

Construction for the purposes of this activity means:

> **any building or engineering work, which includes the repair, alteration, maintenance or improvement of an existing work and preparatory or landscaping works.**

Land reclamation is only permissible under this exemption when it's an integral part of a construction activity.

What type of work can you do?

Example work includes:

- using crushed bricks, concrete, rocks and aggregate to create a noise bund around a new development and then using soil to landscape it to enable grass to grow
- using road planings and rubble to build a track, road or car park
- using wood chip to construct a track, path or bridleway
- bringing in some soil from another place for use in landscaping at a housing development.

Where can you carry out this activity?

This can be done at any place that can comply with the environmental controls listed in the main limits and conditions.

What can't you do?

You **can't**:

- treat waste under this exemption to make it suitable for use
- dispose of waste under this exemption (you can only use waste types that are suitable for use and you should be able to justify the amount of waste that you use)
- use this exemption for land reclamation or disposal in a landfill (you must read the guidance on disposal vs. recovery RGN13: *Defining waste recovery: Permanent deposit of waste on land* and make sure that you are using the waste for recovery only)
- register this exemption more than once at any one place during the three-year period from first registration
- de-register this exemption and then re-register it at the same place within a three-year period
- store the waste for more than 12 months prior to use.

What are the significant limits?

Each table *(on the following pages)* lists the waste types and quantities that can be used over a three-year period from the date of registering the exemption.

You can use up to:

- 5,000 tonnes of any single waste stream or any combination of wastes in Table 1
- 1,000 tonnes of any single waste stream or any combination of wastes in Table 2
- 50,000 tonnes of any single waste stream or any combination of wastes in Table 3.

What are the significant conditions?

You may use a combination of wastes from Tables 1, 2 and 3 provided you do not exceed the limits for each table. Waste can't be stored for longer than 12 months prior to use.

There are three specific conditions to this exemption relating to certain wastes. These are outlined below and also in the relevant section of *What waste can be used under this exemption?*

1. Any person or company can use up to 1,000 tonnes of dredging spoil for any construction (within the 1,000 tonnes total for wastes from Table 2). Exception is made for the Environment Agency and other statutory authorities carrying out land drainage functions under the Land Drainage Act 1991, the Water Resources Act 1991 or the Environment Act 2021. These organisations may use up to 5,000 tonnes of dredging spoil for drainage work (within the 5,000 tonnes total for wastes from Table 1).
2. You can use 1,000 tonnes of wood chip (or similar waste) or road planings to construct tracks, paths, bridleways or car parks only (within the 1,000 tonnes total for wastes from Table 2). The waste must be processed into chipped form prior to use.
3. If you are constructing a road you can use 50,000 tonnes of road planings and road sub-base. The road should be constructed to a specific engineering standard and have a sealed surface in order to qualify for the higher limit.

What waste can be used under this exemption?

Table 1

You can use up to 5,000 tonnes in total of the wastes below for any construction activity.

Codes	Waste types
01 01 02	Waste from mineral non-metalliferous excavation.
01 04 08	Waste gravel and crushed rock other than those mentioned in 01 04 07*.
01 04 09	Waste sand and clays.
02 02 02	Shellfish shells from which the soft tissue or flesh has been removed only.
10 12 08	Waste ceramics, bricks, tiles and construction products (after thermal processing).
10 13 14	Waste concrete and concrete sludge.
17 01 01	Concrete.
17 01 02	Bricks.
17 01 03	Tiles and ceramics.
17 01 07	Mixtures of concrete, bricks, tiles and ceramics other than those mentioned in 17 01 06*.
17 05 08	Track ballast other than those mentioned in 17 05 07*.
19 12 05	Glass.
19 12 09	Minerals (for example sand and stones).
19 12 12	Aggregates only.

Within the 5,000 tonnes total for use of wastes in Table 1, you can only use the waste below for drainage work carried out for the purposes of the Land Drainage Act(1), the Water Resources Act or the Environment Act. This is work that can only be carried out by drainage authorities (such as Inland Drainage Boards, Local Authorities or environment agencies).

Codes	Waste types
17 05 06	Dredging spoil other than those mentioned in 17 05 05*.

Table 2

You can use up to 1,000 tonnes in total of the wastes below for construction purposes.

Codes	Waste types
02 03 99, 02 04 01	Soil from cleaning and washing fruit and vegetables only.
17 05 04	Soil and stones other than those mentioned in 17 05 03*.
17 05 06	Dredging spoil other than those mentioned in 17 05 05*.
19 13 02	Solid wastes from soil remediation other than those mentioned in 19 13 01*.
20 02 02	Soil and stones.

Within the 1,000 tonnes total for use of wastes from Table 2, you can only use the waste below for the construction of tracks, paths, bridleways or car parks. The waste must be processed into chipped form prior to use.

Codes	Waste types
17 03 02	Bituminous mixtures other than those mentioned in 17 03 01*.
02 01 03	Plant tissue waste.
03 01 01, 03 03 01	Untreated waste bark, cork and wood only.
03 01 05	Untreated wood including sawdust, shavings and cuttings from untreated wood only.
17 02 01	Untreated wood only.
19 12 07	Untreated wood other than those mentioned in 19 12 06* only.
20 01 38	Untreated wood other than those mentioned in 20 01 37* only.

** For waste codes beginning with 17 refer to Appendix A; for all other codes refer directly to the European Waste Catalogue.*

Table 3

You can use up to 50,000 tonnes in total of the wastes below only for the construction of roads. The roads should be constructed to a specific engineering standard and have a sealed surface in order to qualify for this larger limit.

Codes	Waste types
17 03 02	Bituminous mixtures other than those mentioned in 17 03 01*.
17 05 04	Road sub-base only.

** Refer to Appendix A.*

Exemption T5 – Screening and blending of waste

This exemption allows temporary small-scale treatment of wastes to produce an aggregate or a soil at a place (such as a construction or demolition site).

What type of work does this cover?

Example work includes:

- screening of soils on a demolition site to remove wood and rubble before sending the soils to a construction site for reuse
- blending of soils and compost that has been produced under an exemption on a construction site to produce a better soil for landscaping works on that site
- crushing wastes (except bricks, tiles and concrete) prior to screening or blending
- grading of waste concrete after crushing to produce a required type of aggregate.

Where can this activity be carried out?

You can treat waste on the site where it is:

- to be used (for example, on a construction site)
- produced (for example, on a demolition site).

What can't you do?

You **can't**:

- import waste, treat it and then export it elsewhere*
- treat waste where the main purpose is disposal to landfill or incineration
- crush waste tiles, bricks or concrete (this comes under a T7 exemption (England and Wales) or Paragraph 19 exemption (Scotland and Northern Ireland), which must be registered with the Local Authority)
- treat hazardous waste.

** This applies even if the treated aggregate meets the quality protocol standard and will no longer be considered as waste.*

What are the significant limits?

- You can store or treat up to 50,000 tonnes of bituminous mixtures for making road stone over a three-year period from the date of registering the exemption.
- You can store or treat up to 5,000 tonnes of other wastes *(listed on the following page)* over a three-year period from the date of registering the exemption.
- Waste can only be stored for up to 12 months.

What are the significant conditions?

Treatment can only be carried out at the place where the waste is to be used or where the waste is produced. This applies even if the resultant material is no longer considered to be waste.

What else do you need to know?

When you have treated the waste the options available are:

- if the treated waste meets the requirements of a waste quality protocol (the quality protocols – WRAP) then it will no longer be considered a waste
- use the treated waste, subject to the significant conditions above, under a use exemption or environmental permit.

What waste can be treated under this exemption?

Codes	Waste types
01 04 08	Waste gravel and crushed rocks other than those mentioned in 01 04 07.*
01 04 09	Waste sand and clays.
02 02 02	Shellfish shells from which the soft tissue or flesh have been removed only.
03 01 01	Untreated waste bark and cork only.
03 03 01	Untreated waste bark and wood.
10 01 01	Bottom ash, slag and boiler dust (excluding boiler dust mentioned in 10 01 04*).
10 01 15	Bottom ash, slag and boiler dust from co-incineration other than those mentioned in 10 01 14.*
17 01 01	Concrete.
17 01 02	Bricks.
17 01 03	Tiles and ceramics.
17 01 07	Mixtures of concrete, bricks, tiles and ceramics other than those mentioned in 17 01 06.*
17 02 01	Untreated wood only.
17 03 02	Bituminous mixtures other than those mentioned in 17 03 01.*
17 05 04	Soil and stones other than those mentioned in 17 05 03.*
17 05 06	Dredging spoil other than those mentioned in 17 05 05.*
17 05 08	Track ballast other than those mentioned in 17 05 07.*
19 05 99	Compost produced pursuant to a treatment described in the paragraphs numbered T23 or T26 of Chapter 2 only.
19 12 05	Glass.
19 12 09	Aggregates only.
19 12 12	Gypsum recovered from construction materials only.
19 13 02	Solid wastes from soil remediation other than those mentioned in 19 13 01.*
19 13 04	Sludges from remediation other than those mentioned in 19 13 03.*
20 02 02	Soil and stones.

** For waste codes beginning with 17 refer to Appendix A; for all other codes refer directly to the European Waste Catalogue.*

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Scan the QR code to complete a brief survey, and share your feedback on the course.

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SITE SAFETY PLUS

Construction site health and safety
E: Environment

This section is dedicated to the management of environmental issues related to the construction industry and includes information to raise energy efficiency awareness.

It explains what the law requires, what needs to be done and practical advice on how to implement and maintain the controls needed to manage the basics on site, as well as in more challenging environments.

This publication is also the official reference material for the Site Safety Plus *Site environmental awareness training scheme* (SEATS®), a one-day course for supervisors.

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